Defining Group Homology and Cohomology

Revised Notes 1

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Abstract

These notes cover the first three meetings of the Fall 2026 Geometry and Topology Seminar. Starting from left and right modules and resolutions, we construct left and right derived functors and prove independence from the chosen resolutions. Tensor and Hom yield Tor and Ext, and hence group homology and cohomology through coinvariants and invariants. Calculations for the infinite cyclic group illustrate both constructions. The bar resolution provides general formulas, and an appendix proves the existence of injective resolutions.

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1 Left Modules, Presentations, and Free Resolutions

1.1 Rings and left modules

Ring. A *unital ring* is an abelian group $(R, +, 0)$ with a multiplication and an element 1_R such that, for all $r, s, t \in R$,

$$(rs)t = r(st), \qquad r(s+t) = rs + rt, \qquad (r+s)t = rt + st, \qquad 1_R r = r = r 1_R.$$

Throughout, R is a unital ring, not necessarily commutative; $1_R = 0$ is allowed.

Left module. A *left R-module* is an abelian group $(M, +, 0)$ with an action $R \times M \to M$, $(r, m) \mapsto rm$, satisfying

$$r(m+m') = rm + rm', \quad (r+r')m = rm + r'm,$$
$$(rr')m = r(r'm), \qquad 1_R m = m$$

for all $r, r' \in R$ and $m, m' \in M$. We write R-**Mod** for left R-modules. Unless a right action is specified, R-linear means left R-linear.

Submodule. A *submodule* $K \subseteq M$ is an additive subgroup satisfying $rk \in K$ for all $r \in R$ and $k \in K$.

Quotient left module. For a submodule $K \subseteq M$, the set of cosets $M/K = \{m + K : m \in M\}$ is a left R-module with

$$(m+K) + (m'+K) = (m+m') + K, \qquad r(m+K) = rm + K.$$

The quotient map $q : M \to M/K$, $q(m) = m + K$, is surjective.

Left module homomorphism. A map $f : M \to N$ is *R-linear* if

$$f(m+m') = f(m) + f(m'), \qquad f(rm) = r f(m).$$

An R-linear map is a *monomorphism* if it is injective, an *epimorphism* if it is surjective, and an *isomorphism* if it is bijective.

Kernel and image. For an R-linear map $f : M \to N$,

$$\ker f = \{m \in M : f(m) = 0\}, \qquad \operatorname{im} f = \{f(m) : m \in M\}$$

are submodules of M and N, respectively.

First isomorphism theorem. Every R-linear map $f : M \to N$ induces an isomorphism

$$M/\ker f \xrightarrow{\cong} \operatorname{im} f, \qquad m + \ker f \longmapsto f(m).$$

Proof. Since $f(m) = f(m')$ holds exactly when $m + \ker f = m' + \ker f$, the displayed map is well defined and injective. It is R-linear by linearity of f and surjective by the definition of $\operatorname{im} f$, hence an isomorphism. □

1.2 Free left modules and presentations

Definition by the universal property. A *free left R-module on a set S* is a pair (F, ι), where F is a left R-module and $\iota : S \to F$ is a function, such that for every left R-module N and every function $a : S \to N$, there is a unique R-linear map $\widetilde{a} : F \to N$ with

$$\widetilde{a} \circ \iota = a.$$

A left R-module is *free* if it is free on some set.

Existence. Set

$$F_R(S) = \bigoplus_{s \in S} Re_s.$$

Here each Re_s is a copy of the left module R, labelled by s, and the *direct sum* means

$$\bigoplus_{s \in S} Re_s = \{(r_s)_{s \in S} : r_s \in R,\ r_s = 0 \text{ for all but finitely many } s\},$$

with coordinatewise addition and left scalar multiplication. The vector e_s has entry 1_R at s and 0 elsewhere. Thus every element has a unique expression $\sum_{s \in S} r_s e_s$ with finitely many nonzero coefficients; $(e_s)_{s \in S}$ is its *basis*.

Put $\iota(s) = e_s$. For a function $a : S \to N$, the formula

$$\widetilde{a}\left(\sum_s r_s e_s\right) = \sum_s r_s a(s)$$

defines an R-linear map extending a. Every such extension must satisfy this formula, proving the universal property. □

Uniqueness. If (F, ι) and (F', ι') are free on the same set S, their universal properties give unique R-linear maps

$$u : F \to F', \quad u\iota = \iota', \qquad v : F' \to F, \quad v\iota' = \iota.$$

Uniqueness gives $vu = \mathrm{id}_F$ and $uv = \mathrm{id}_{F'}$. Hence there is a unique isomorphism $u : F \to F'$ satisfying $u\iota = \iota'$. □

Generators and relations. A subset $S \subseteq M$ *generates* M if every element of M is a finite sum of elements rs with $r \in R$ and $s \in S$. Choose such an S; taking $S = M$ is always possible. The universal property gives a surjection

$$\varepsilon : F_0 = F_R(S) \longrightarrow M, \qquad \sum_s r_s e_s \longmapsto \sum_s r_s s.$$

With $K_0 = \ker \varepsilon$, the first isomorphism theorem gives

$$M \cong F_0/K_0, \qquad K_0 = \left\{\sum_s r_s e_s : \sum_s r_s s = 0 \text{ in } M\right\}.$$

The basis elements e_s are formal *generators*; the elements of K_0 are *relations* among their images in M. This is called a *presentation* of M.

1.3 Exact sequences and free resolutions

From a quotient to a short exact sequence. Write $M \cong F_0/K_0$ from Section 1.2 as

$$0 \longrightarrow K_0 \xrightarrow{i_0} F_0 \xrightarrow{\varepsilon} M \longrightarrow 0,$$

where i_0 is inclusion, ε is surjective, and $\operatorname{im} i_0 = K_0 = \ker \varepsilon$. This is a *short exact sequence.*

In general, a sequence of left R-modules and R-linear maps

$$\cdots \longrightarrow A_{n+1} \xrightarrow{f_{n+1}} A_n \xrightarrow{f_n} A_{n-1} \longrightarrow \cdots$$

is *exact at* A_n if $\operatorname{im} f_{n+1} = \ker f_n$, and *exact* if this holds at every term having an incoming and an outgoing map. In particular,

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is short exact precisely when f is injective, $\operatorname{im} f = \ker g$, and g is surjective.

Relations among relations. Choose a generating set $T \subseteq K_0$. The resulting free left R-module $F_1 = F_R(T)$ maps onto K_0 by $p_1(e_t) = t$. Put $K_1 = \ker p_1$ and let $i_1 : K_1 \hookrightarrow F_1$ be inclusion. This gives a second short exact sequence

$$0 \longrightarrow K_1 \xrightarrow{i_1} F_1 \xrightarrow{p_1} K_0 \longrightarrow 0.$$

Define $d_1 = i_0 p_1 : F_1 \to F_0$. Splicing the two sequences gives

$$0 \longrightarrow K_1 \xrightarrow{i_1} F_1 \xrightarrow{d_1} F_0 \xrightarrow{\varepsilon} M \longrightarrow 0,$$

which is exact since $\ker d_1 = K_1$ and $\operatorname{im} d_1 = K_0$.

The elements $t \in T$ are chosen relations in F_0. An element $\sum_t a_t e_t \in F_1$ belongs to K_1 exactly when

$$\sum_t a_t t = 0 \quad \text{in } F_0.$$

Thus K_1 consists of the *relations among the chosen relations.*

Iterating to get a free resolution. For each $n \geq 1$, choose a free left R-module F_n and a surjection $p_n : F_n \to K_{n-1}$, put $K_n = \ker p_n$, and let $i_n : K_n \hookrightarrow F_n$ be inclusion. Set $d_n = i_{n-1} p_n$. Then

$$\cdots \longrightarrow F_2 \xrightarrow{d_2} F_1 \xrightarrow{d_1} F_0 \xrightarrow{\varepsilon} M \longrightarrow 0$$

is exact: $\operatorname{im} d_1 = \ker \varepsilon$ and $\operatorname{im} d_{n+1} = K_n = \ker d_n$ for every $n \geq 1$.

An exact sequence of this form with every F_n free is called a *free resolution* of M; ε is its *augmentation.* The construction proves that every left R-module has a free resolution. □

Higher free left R-modules encode successive relations among relations.

Removing the augmentation to get a chain complex. Delete M and ε from the resolution, leaving

$$F_\bullet : \qquad \cdots \longrightarrow F_2 \xrightarrow{d_2} F_1 \xrightarrow{d_1} F_0 \longrightarrow 0.$$

Its consecutive maps compose to zero.

In general, a *chain complex* of left R-modules is a family $(C_n)_{n\in\mathbb{Z}}$ with R-linear maps $d_n : C_n \to C_{n-1}$, called *differentials*, such that

$$d_n d_{n+1} = 0 \qquad \text{for every } n.$$

Equivalently, $\operatorname{im} d_{n+1} \subseteq \ker d_n$; equality is not required.

The *nth homology* of $C_\bullet$ is the quotient left R-module

$$H_n(C_\bullet) = \ker d_n / \operatorname{im} d_{n+1}.$$

Its scalar action is $r[x] = [rx]$. Elements of $\ker d_n$ are *n-cycles*; elements of $\operatorname{im} d_{n+1}$ are *n-boundaries*. Thus $H_n(C_\bullet) = 0$ precisely when the complex is exact at C_n.

The complex $F_\bullet$ has $F_n = 0$ for $n < 0$ and $d_0 = 0$. In terms of homology, the free resolution gives

$$H_0(F_\bullet) = F_0/\operatorname{im} d_1 \cong M, \qquad H_n(F_\bullet) = 0 \quad (n \neq 0).$$

Example 1: one generator and one relation. Let $n \geq 2$ and regard $M = \mathbb{Z}/n\mathbb{Z}$ as a left $\mathbb{Z}$-module. It is generated by $\overline{1}$ with the relation $n\overline{1} = 0$. The standard free resolution is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \xrightarrow{\varepsilon} \mathbb{Z}/n\mathbb{Z} \longrightarrow 0, \qquad \varepsilon(a) = \overline{a}.$$

Multiplication by n is injective and its image is $n\mathbb{Z} = \ker \varepsilon$, so the sequence is exact.

Example 2: a redundant relation. Keep the same generator, but choose both relations $n\overline{1} = 0$ and $2n\overline{1} = 0$. Thus $F_0 = \mathbb{Z}$, $F_1 = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 = \mathbb{Z}^2$, and

$$d_1 : F_1 \longrightarrow F_0, \qquad d_1(a, b) = na + 2nb = n(a + 2b).$$

The second relation is redundant: $2n = 2 \cdot n$ in F_0. We have

$$\ker d_1 = \{(a, b) \in \mathbb{Z}^2 : a + 2b = 0\} = \mathbb{Z}(-2, 1).$$

The nonzero element $-2e_1 + e_2 \in F_1$ records the identity $-2n + 2n = 0$ in F_0: a *relation among the chosen relations*. The map

$$d_2 : \mathbb{Z} \longrightarrow \mathbb{Z}^2, \qquad t \longmapsto (-2t, t)$$

is injective with image $\ker d_1$. Since $\operatorname{im} d_1 = n\mathbb{Z} = \ker \varepsilon$, we obtain the free resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d_2} \mathbb{Z}^2 \xrightarrow{d_1} \mathbb{Z} \xrightarrow{\varepsilon} \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

The additional term records the redundancy of the chosen relations.

A free resolution thus unravels a left R-module, layer by layer: generators, relations, relations among relations, and so on.

2 Projective Resolutions and Left Derived Functors

2.1 Projective left modules

Definition by the lifting property. A left R-module P is *projective* if, for every epimorphism $q : A \to B$ of left R-modules and every R-linear map $f : P \to B$, there exists an R-linear map $\widetilde{f} : P \to A$ such that

$$q \circ \widetilde{f} = f.$$

The map $\widetilde{f}$ is a *lift* of f through q. The lift need not be unique.

Proposition. Every free left R-module is projective.

Proof. Let F be free with basis $(e_s)_{s\in S}$. Given an epimorphism $q : A \to B$ and an R-linear map $f : F \to B$, choose $a_s \in A$ with $q(a_s) = f(e_s)$ for each $s \in S$. By the universal property of F, the assignment $e_s \mapsto a_s$ extends uniquely to an R-linear map $\widetilde{f} : F \to A$. The maps $q \circ \widetilde{f}$ and f agree on every basis element, so $q \circ \widetilde{f} = f$. This proves projectivity. The extension is unique for the chosen a_s, but the choices themselves need not be unique. □

Since every left R-module is a quotient of a free left R-module, every left R-module admits an epimorphism from a projective left R-module. Thus there are *enough projective left modules*. Freeness is strictly stronger than projectivity in general: every free left R-module is projective, but a projective left R-module need not be free.

We introduce projective left R-modules because their lifting property is useful in later proofs and dualizes to the extension property of injective left R-modules. For most explicit computations in these notes, we will use free left or right R-modules, as appropriate, and free resolutions: they provide concrete generators and relations, whereas identifying suitable projective or injective left R-modules, and assembling them into resolutions, can be difficult.

2.2 Projective resolutions

Definition and notation. A *projective resolution* of a left R-module M, written $P_\bullet \xrightarrow{\varepsilon} M$, is an exact sequence

$$\cdots \longrightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \longrightarrow 0$$

in which every P_n is projective. The map $\varepsilon : P_0 \to M$ is the *augmentation*.

The notation $P_\bullet$ alone denotes the underlying chain complex

$$\cdots \longrightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \longrightarrow 0,$$

with $P_n = 0$ for $n < 0$ and $d_0 = 0$. It does not include M; the full augmented exact sequence does. The augmentation identifies $H_0(P_\bullet)$ with M, and $H_n(P_\bullet) = 0$ for $n > 0$.

Existence. Every left R-module has a free resolution. Since every free left R-module is projective, this is also a projective resolution. □

Towards uniqueness. The two resolutions in Section 1.3 have different lengths and are not isomorphic degree by degree. We will establish uniqueness up to *chain homotopy equivalence*.

The preparation parallels topology: continuous maps between spaces, homotopies between maps, and homotopy equivalences between spaces.

Chain maps. Let $C_\bullet$ and $D_\bullet$ be chain complexes of left R-modules. A *chain map* $f : C_\bullet \to D_\bullet$ is a family of R-linear maps $f_n : C_n \to D_n$ such that

$$d_n^D \circ f_n = f_{n-1} \circ d_n^C \qquad \text{for every } n.$$

Equivalently, the following squares commute:

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & C_{n+1} & \xrightarrow{d_{n+1}^C} & C_n & \xrightarrow{d_n^C} & C_{n-1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle f_{n+1}} & & \downarrow{\scriptstyle f_n} & & \downarrow{\scriptstyle f_{n-1}} & & \\
\cdots & \longrightarrow & D_{n+1} & \xrightarrow[d_{n+1}^D]{} & D_n & \xrightarrow[d_n^D]{} & D_{n-1} & \longrightarrow & \cdots
\end{array}$$

Composition and identity maps are defined degree by degree.

Proposition. Every chain map $f : C_\bullet \to D_\bullet$ canonically induces an R-linear map

$$H_n(f) : H_n(C_\bullet) \longrightarrow H_n(D_\bullet), \qquad [x] \longmapsto [f_n(x)]$$

for every n. These maps satisfy $H_n(g \circ f) = H_n(g) \circ H_n(f)$ for composable chain maps, and $H_n(\mathrm{id}_{C_\bullet}) = \mathrm{id}_{H_n(C_\bullet)}$.

Proof. The chain map condition gives

$$f_n(\ker d_n^C) \subseteq \ker d_n^D, \qquad f_n(\operatorname{im} d_{n+1}^C) \subseteq \operatorname{im} d_{n+1}^D.$$

Thus the formula is well-defined on the quotient and is R-linear. Compatibility with composition and identities follows from the formula. □

Chain homotopies. Let $f, g : C_\bullet \to D_\bullet$ be chain maps. A *chain homotopy* from f to g is a family of R-linear maps $s_n : C_n \to D_{n+1}$ such that

$$f_n - g_n = d_{n+1}^D \circ s_n + s_{n-1} \circ d_n^C \qquad \text{for every } n.$$

In the following diagram, the middle vertical map is the sum of the two composites from C_n to D_n using a dashed arrow:

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & C_{n+1} & \xrightarrow{d_{n+1}^C} & C_n & \xrightarrow{d_n^C} & C_{n-1} & \longrightarrow & \cdots \\
 & & & \swarrow{\scriptstyle s_n} & \downarrow{\scriptstyle f_n - g_n} & \searrow{\scriptstyle s_{n-1}} & & & \\
\cdots & \longrightarrow & D_{n+1} & \xrightarrow[d_{n+1}^D]{} & D_n & \xrightarrow[d_n^D]{} & D_{n-1} & \longrightarrow & \cdots
\end{array}$$

If such a family exists, f and g are *chain homotopic*, written $f \simeq g$. For complexes concentrated in nonnegative degrees, $s_{-1} = 0$.

Proposition. Chain homotopic maps $f, g : C_\bullet \to D_\bullet$ induce the same maps on homology: $H_n(f) = H_n(g)$ for every n.

Proof. If $x \in \ker d_n^C$, the homotopy identity gives

$$f_n(x) - g_n(x) = d_{n+1}^D s_n(x) \in \operatorname{im} d_{n+1}^D.$$

Hence $[f_n(x)] = [g_n(x)]$. □

Chain homotopy equivalences. A chain map $f : C_\bullet \to D_\bullet$ is a *chain homotopy equivalence* if there is a chain map $g : D_\bullet \to C_\bullet$ such that

$$g \circ f \simeq \mathrm{id}_{C_\bullet}, \qquad f \circ g \simeq \mathrm{id}_{D_\bullet}.$$

The map g is a *homotopy inverse* of f. In this case the complexes are *chain homotopy equivalent*, written $C_\bullet \simeq D_\bullet$.

Proposition. A chain homotopy equivalence $f : C_\bullet \to D_\bullet$ induces an isomorphism $H_n(f) : H_n(C_\bullet) \to H_n(D_\bullet)$ for every n.

Proof. Let g be a homotopy inverse of f. The preceding propositions give

$$H_n(g) \circ H_n(f) = H_n(g \circ f) = H_n(\mathrm{id}_{C_\bullet}) = \mathrm{id}_{H_n(C_\bullet)},$$
$$H_n(f) \circ H_n(g) = H_n(f \circ g) = H_n(\mathrm{id}_{D_\bullet}) = \mathrm{id}_{H_n(D_\bullet)}.$$

The middle equality in each line uses $g \circ f \simeq \mathrm{id}_{C_\bullet}$ and $f \circ g \simeq \mathrm{id}_{D_\bullet}$: chain homotopic maps induce the same maps on homology. The first and last equalities follow from compatibility with composition and identity maps, respectively. Thus $H_n(g)$ is the inverse of $H_n(f)$. □

Remark. For later applications, merely inducing isomorphisms on homology is too weak: a chain map can lose this property after a non-exact additive covariant functor is applied degree by degree. Such a functor does preserve chain homotopies and chain homotopy equivalences, because it preserves sums and compositions. We will therefore apply the functor to homotopy equivalent resolutions and then use the preceding proposition to compare the resulting homology.

Example: comparing the two resolutions. Let $P_\bullet$ and $Q_\bullet$ be the underlying complexes of the standard and redundant-relation resolutions of $\mathbb{Z}/n\mathbb{Z}$ in Section 1.3, with $n \geq 2$. The diagrams below show the full augmented resolutions, with $\mathbb{Z}/n\mathbb{Z}$ in degree -1 and zeros at both ends. The maps f and g are extended by the identity in degree -1. Recall that

$$d_2(t) = (-2t, t), \qquad d_1(a, b) = n(a + 2b).$$

Define $f : P_\bullet \to Q_\bullet$ by $f_1(x) = (x, 0)$ and $f_0 = \mathrm{id}$, with all other degree components zero. Extend it to the augmented complexes by $f_{-1} = \mathrm{id}$:

$$\begin{array}{ccccccccccccc}
0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\times n} & \mathbb{Z} & \xrightarrow{\varepsilon} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \\
\downarrow & & \downarrow{\scriptstyle f_2=0} & & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_0=\mathrm{id}} & & \downarrow{\scriptstyle f_{-1}=\mathrm{id}} & & \downarrow \\
0 & \longrightarrow & \mathbb{Z} & \xrightarrow[d_2]{} & \mathbb{Z}^2 & \xrightarrow[d_1]{} & \mathbb{Z} & \xrightarrow[\varepsilon]{} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0
\end{array}$$

In the other direction, define $g : Q_\bullet \to P_\bullet$ by $g_1(a, b) = a + 2b$ and $g_0 = \mathrm{id}$, with all other degree

components zero. Extend it to the augmented complexes by $g_{-1} = \mathrm{id}$:

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & \mathbb{Z} & \xrightarrow{d_2} & \mathbb{Z}^2 & \xrightarrow{d_1} & \mathbb{Z} & \xrightarrow{\varepsilon} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \\
\downarrow & & \downarrow{\scriptstyle g_2=0} & & \downarrow{\scriptstyle g_1} & & \downarrow{\scriptstyle g_0=\mathrm{id}} & & \downarrow{\scriptstyle g_{-1}=\mathrm{id}} & & \downarrow \\
0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \xrightarrow[\times n]{} & \mathbb{Z} & \xrightarrow[\varepsilon]{} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0
\end{array}$$

The identities $d_1 f_1(x) = nx$, $n g_1(a,b) = d_1(a,b)$, and $g_1 d_2(t) = 0$ verify that the squares commute. Both maps respect the common augmentation $\varepsilon : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$.

The composite gf. Since $g_1 f_1(x) = x$, we have $gf = \mathrm{id}_{P_\bullet}$:

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\times n} & \mathbb{Z} & \xrightarrow{\varepsilon} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \\
\downarrow & & \downarrow{\scriptstyle \mathrm{id}_0} & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow \\
0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \xrightarrow[\times n]{} & \mathbb{Z} & \xrightarrow[\varepsilon]{} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0
\end{array}$$

and $\mathrm{id}_{P_\bullet} \simeq \mathrm{id}_{P_\bullet}$ trivially.

The composite fg. Put $e(a,b) = (a + 2b, 0)$. Then $(fg)_2 = 0$, $(fg)_1 = e$, and $(fg)_0 = \mathrm{id}$:

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & \mathbb{Z} & \xrightarrow{d_2} & \mathbb{Z}^2 & \xrightarrow{d_1} & \mathbb{Z} & \xrightarrow{\varepsilon} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \\
\downarrow & & \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle e} & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow \\
0 & \longrightarrow & \mathbb{Z} & \xrightarrow[d_2]{} & \mathbb{Z}^2 & \xrightarrow[d_1]{} & \mathbb{Z} & \xrightarrow[\varepsilon]{} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0
\end{array}$$

A homotopy from fg to the identity. Define $s_1 : Q_1 \to Q_2$ by $s_1(a,b) = -b$, and set $s_j = 0$ for $j \neq 1$, including $s_{-1} = 0$. The dashed arrow is the only nonzero homotopy component; the vertical maps extend $fg - \mathrm{id}_{Q_\bullet}$ by zero in degree -1:

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & \mathbb{Z} & \xrightarrow{d_2} & \mathbb{Z}^2 & \xrightarrow{d_1} & \mathbb{Z} & \xrightarrow{\varepsilon} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \\
\downarrow & & \downarrow{\scriptstyle -\mathrm{id}} & \overset{s_1}{\swarrow} & \downarrow{\scriptstyle e-\mathrm{id}} & & \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle 0} & & \downarrow \\
0 & \longrightarrow & \mathbb{Z} & \xrightarrow[d_2]{} & \mathbb{Z}^2 & \xrightarrow[d_1]{} & \mathbb{Z} & \xrightarrow[\varepsilon]{} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0
\end{array}$$

Indeed,

$$d_2 s_1(a,b) = (2b, -b) = e(a,b) - (a,b),$$
$$s_1 d_2(t) = -t = (fg)_2(t) - t.$$

In degree 0, both sides of the homotopy identity are zero. The degree--1 identity is also zero since $s_{-1} = 0$. Thus $fg - \mathrm{id}_{Q_\bullet} = ds + sd$, so $fg \simeq \mathrm{id}_{Q_\bullet}$. Together with $gf = \mathrm{id}_{P_\bullet}$, this proves that f and g are homotopy inverses and $P_\bullet \simeq Q_\bullet$. The extra relation in $Q_\bullet$ is therefore removed up to chain homotopy.

The same maps and homotopy remain valid after deleting the augmentation term $\mathbb{Z}/n\mathbb{Z}$, since $s_{-1} = 0$. The same construction extends to arbitrary projective resolutions. The proof makes the role of projectivity explicit: exactness of the target resolution supplies surjections onto kernels, and projectivity of the source terms supplies lifts through these surjections.

Theorem (Comparison of projective resolutions). Let $P_\bullet \xrightarrow{\varepsilon_P} M$ and $Q_\bullet \xrightarrow{\varepsilon_Q} N$ be projective resolutions of left R-modules. In the augmented complexes, M and N occupy degree -1, and all lower terms are zero. Their degree-0 differentials are the augmentations; the underlying complexes $P_\bullet, Q_\bullet$ still have $d_0^P = d_0^Q = 0$. For every R-linear map $u : M \to N$, the following hold.

1. *Existence of a lift.* There is a chain map $f : P_\bullet \to Q_\bullet$ satisfying

$$\varepsilon_Q f_0 = u\varepsilon_P.$$

   Together, f and u define a chain map of augmented complexes

$$\widetilde{u} = (f, u) : (P_\bullet \xrightarrow{\varepsilon_P} M) \longrightarrow (Q_\bullet \xrightarrow{\varepsilon_Q} N),$$

   called a *lift of* u. Its component is f_n in degree $n \geq 0$ and u in degree -1.

2. *Uniqueness up to chain homotopy.* Any two lifts $\widetilde{u} = (f, u)$ and $\widetilde{u}' = (f', u)$ are chain homotopic. The homotopy can be chosen in the form

$$\widetilde{s} = (s, 0), \qquad \widetilde{s}_n = s_n \quad (n \geq 0), \qquad \widetilde{s}_{-1} : M \to Q_0 \text{ is zero},$$

   where $s = (s_n)_{n \geq 0}$ is a chain homotopy from f to f' on the underlying complexes $P_\bullet, Q_\bullet$.

Proof (Existence). Start with the two augmented resolutions and the given map u:

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & P_2 & \xrightarrow{d_2^P} & P_1 & \xrightarrow{d_1^P} & P_0 & \xrightarrow{\varepsilon_P} & M & \longrightarrow & 0 \\
 & & & & & & & & \downarrow{\scriptstyle u} & & \\
\cdots & \longrightarrow & Q_2 & \xrightarrow{d_2^Q} & Q_1 & \xrightarrow{d_1^Q} & Q_0 & \xrightarrow{\varepsilon_Q} & N & \longrightarrow & 0.
\end{array}$$

We seek maps $f_n : P_n \to Q_n$ extending u to a chain map of these augmented complexes. First, $u\varepsilon_P : P_0 \to N$ is R-linear and $\varepsilon_Q : Q_0 \to N$ is an epimorphism. Projectivity of P_0 gives a lift $f_0 : P_0 \to Q_0$ making the following diagram commute:

$$\begin{array}{ccc}
P_0 & & \\
{\scriptstyle f_0}\downarrow & \searrow^{u\varepsilon_P} & \\
Q_0 & \xrightarrow[\varepsilon_Q]{} & N.
\end{array}$$

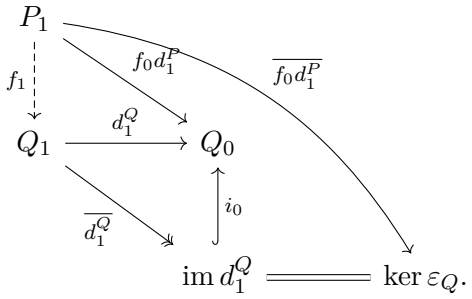
Thus $\varepsilon_Q f_0 = u\varepsilon_P$, which is the chain map condition at the augmentation.

Next, consider $f_0 d_1^P : P_1 \to Q_0$. We check that

$$\varepsilon_Q f_0 d_1^P = u\varepsilon_P d_1^P = 0.$$

Thus its image lies in $\ker \varepsilon_Q$, giving a map $\overline{f_0 d_1^P} : P_1 \to \ker \varepsilon_Q$. Exactness identifies $\ker \varepsilon_Q = \operatorname{im} d_1^Q$. Write $i_0 : \operatorname{im} d_1^Q \hookrightarrow Q_0$ for the inclusion and $\overline{d_1^Q} : Q_1 \twoheadrightarrow \operatorname{im} d_1^Q$ for the induced epimorphism. Both

bars change only the codomain of the original map. Projectivity of P_1 supplies the dashed lift:



Under the displayed identification, $\overline{d_1^Q} f_1 = \overline{f_0 d_1^P}$. Composing with i_0 gives

$$d_1^Q f_1 = i_0 \overline{d_1^Q} f_1 = i_0 \overline{f_0 d_1^P} = f_0 d_1^P,$$

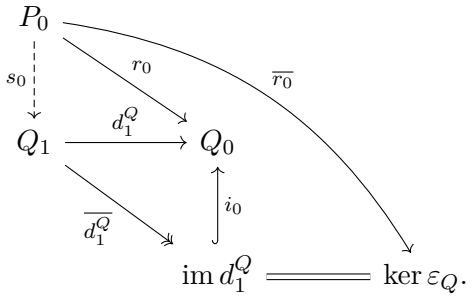
which is the chain map condition in degree 1. Repeating this construction gives all f_n, with $d_n^Q f_n = f_{n-1} d_n^P$ for $n \geq 1$. Thus $f = (f_n)$ is a chain map, and $\varepsilon_Q f_0 = u \varepsilon_P$ makes $\widetilde{u} = (f, u)$ the required augmented lift. The formal induction is omitted.

(Uniqueness up to homotopy). Let $\widetilde{u} = (f, u)$ and $\widetilde{u}' = (f', u)$ be two lifts.

Degree 0. Set $s_{-1} = 0$ on the underlying complexes and put $r_0 = f_0 - f_0'$. Since both maps lift u,

$$\varepsilon_Q r_0 = u \varepsilon_P - u \varepsilon_P = 0.$$

Thus r_0 takes values in $\ker \varepsilon_Q = \operatorname{im} d_1^Q$. Projectivity of P_0 gives the dashed lift below; as before, bars change only codomains, and i_0 is the inclusion.



The lift satisfies $\overline{d_1^Q} s_0 = \overline{r_0}$. Composing with i_0 gives

$$f_0 - f_0' = r_0 = d_1^Q s_0,$$

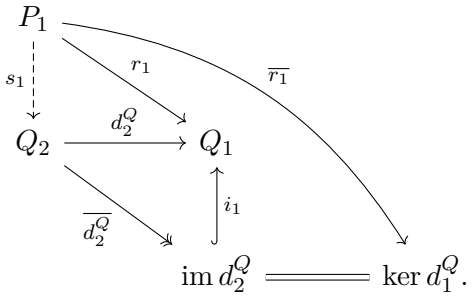
the homotopy identity in degree 0.

Degree 1. Define $r_1 = f_1 - f_1' - s_0 d_1^P : P_1 \to Q_1$. The chain map identities and the degree-0 identity give

$$\begin{aligned} d_1^Q r_1 &= (f_0 - f_0') d_1^P - d_1^Q s_0 d_1^P \\ &= (f_0 - f_0' - d_1^Q s_0) d_1^P = 0. \end{aligned}$$

Hence r_1 takes values in $\ker d_1^Q = \operatorname{im} d_2^Q$. Writing $i_1 : \operatorname{im} d_2^Q \hookrightarrow Q_1$ for the inclusion, projectivity

of P_1 gives the next lift:



Composing $\overline{d_2^Q} s_1 = \overline{r_1}$ with i_1 gives $d_2^Q s_1 = r_1$, so

$$f_1 - f_1' = d_2^Q s_1 + s_0 d_1^P.$$

Degree 2. Put $r_2 = f_2 - f_2' - s_1 d_2^P : P_2 \to Q_2$. This time one extra term remains, but it still vanishes:

$$d_2^Q r_2 = (f_1 - f_1' - d_2^Q s_1) d_2^P = s_0 d_1^P d_2^P = 0.$$

Exactness and projectivity of P_2 again give a lift $s_2 : P_2 \to Q_3$ with $d_3^Q s_2 = r_2$. Continuing in the same way produces all s_n satisfying

$$f_n - f_n' = d_{n+1}^Q s_n + s_{n-1} d_n^P \qquad (n \geq 0).$$

Thus s is a chain homotopy from f to f' on the underlying complexes. The formal induction is omitted.

Extend s to the augmented complexes by setting $\widetilde{s}_{-1} : M \to Q_0$ equal to zero. In degree 0, the additional term $\widetilde{s}_{-1}\varepsilon_P$ vanishes; in degree -1, the homotopy identity is

$$u - u = \varepsilon_Q \widetilde{s}_{-1} = 0.$$

Thus $\widetilde{s} = (s, 0)$ is a chain homotopy from $\widetilde{u}$ to $\widetilde{u}'$. □

Corollary (Uniqueness of projective resolutions). Any two projective resolutions of the same left R-module are chain homotopy equivalent, both as augmented complexes and after deleting the augmentation terms.

Proof. Label two copies of the same left R-module A by $A^{(1)}$ and $A^{(2)}$, and write

$$\widetilde{P}_\bullet = (P_\bullet \xrightarrow{\varepsilon_P} A^{(1)}), \qquad \widetilde{Q}_\bullet = (Q_\bullet \xrightarrow{\varepsilon_Q} A^{(2)}).$$

The map $\mathrm{id}_A : A^{(1)} \to A^{(2)}$ lifts by the comparison theorem, as does $\mathrm{id}_A : A^{(2)} \to A^{(1)}$ in the reverse direction:

$$\widetilde{f} = (f, \mathrm{id}_A) : \widetilde{P}_\bullet \longrightarrow \widetilde{Q}_\bullet,$$
$$\widetilde{g} = (g, \mathrm{id}_A) : \widetilde{Q}_\bullet \longrightarrow \widetilde{P}_\bullet.$$

Here id_A denotes the identity on the underlying left R-module A. The composites satisfy

$$\widetilde{g}\widetilde{f} = (gf, \mathrm{id}_A) \quad \text{lifts } \mathrm{id}_A : A^{(1)} \to A^{(1)},$$
$$\widetilde{f}\widetilde{g} = (fg, \mathrm{id}_A) \quad \text{lifts } \mathrm{id}_A : A^{(2)} \to A^{(2)}.$$

But the identity chain maps

$$\mathrm{id}_{\widetilde{P}_\bullet} = (\mathrm{id}_{P_\bullet}, \mathrm{id}_A), \qquad \mathrm{id}_{\widetilde{Q}_\bullet} = (\mathrm{id}_{Q_\bullet}, \mathrm{id}_A)$$

also lift these respective left R-linear maps. Uniqueness up to homotopy gives

$$\widetilde{g}\widetilde{f} \simeq \mathrm{id}_{\widetilde{P}_\bullet}, \qquad \widetilde{f}\widetilde{g} \simeq \mathrm{id}_{\widetilde{Q}_\bullet}.$$

Thus $\widetilde{P}_\bullet \simeq \widetilde{Q}_\bullet$. The theorem provides both homotopies with degree--1 component zero, so deleting $A^{(1)}$ and $A^{(2)}$ leaves $gf \simeq \mathrm{id}_{P_\bullet}$ and $fg \simeq \mathrm{id}_{Q_\bullet}$. Hence $P_\bullet \simeq Q_\bullet$ as well. □

The strict equality $gf = \mathrm{id}_{P_\bullet}$ in the preceding example is a special feature of the chosen maps; in general, both composites are only required to be homotopic to the respective identities.

Remark. In particular, f induces isomorphisms $H_n(P_\bullet) \cong H_n(Q_\bullet)$. These recover A in degree 0 and zero in positive degrees. As noted earlier, the useful conclusion for later applications is the chain homotopy equivalence: every additive covariant functor F between categories of left modules over possibly different rings preserves it, even when F is not exact. Thus $F(P_\bullet) \simeq F(Q_\bullet)$ and their homology is isomorphic in each degree. However, a homology isomorphism may not be preserved by F alone.

Remark. For computations, we choose a projective resolution as simple as possible. In practice, we use simple free resolutions.

2.3 Left derived functors

Categories. A *category* $\mathcal{C}$ consists of:

- a collection of objects;
- for each pair (X, Y) of objects, a set $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ of morphisms $f : X \to Y$;
- an identity morphism $\mathrm{id}_X : X \to X$ for each object X;
- composition $\mathrm{Hom}_{\mathcal{C}}(Y, Z) \times \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{C}}(X, Z)$, sending (g, f) to $g \circ f$.

These satisfy, whenever the compositions are defined,

$$h \circ (g \circ f) = (h \circ g) \circ f, \qquad \mathrm{id}_Y \circ f = f = f \circ \mathrm{id}_X \quad (f : X \to Y).$$

Covariant functors. A *covariant functor* $F : \mathcal{C} \to \mathcal{D}$ assigns:

- to each object X, an object $F(X)$;
- to each morphism $f : X \to Y$, a morphism $F(f) : F(X) \to F(Y)$.

It must satisfy

$$F(\mathrm{id}_X) = \mathrm{id}_{F(X)}, \qquad F(g \circ f) = F(g) \circ F(f).$$

The categories of interest.

- R-**Mod** has left R-modules as objects and R-linear maps as morphisms.
- **Ab** has abelian groups as objects and group homomorphisms as morphisms.

Identities and composition are the usual ones for maps. Every abelian group has a unique unital left $\mathbb{Z}$-module structure, and its group homomorphisms are exactly the $\mathbb{Z}$-linear maps. Thus **Ab** is canonically isomorphic to $\mathbb{Z}$-**Mod**; we freely identify them. In particular, the preceding definitions of chain complexes, homology, and chain homotopies apply to abelian groups.

Additive functors. In both R-**Mod** and **Ab**, each morphism set is an abelian group under pointwise addition, and composition is bilinear (when treating morphism sets as left $\mathbb{Z}$-modules).

A covariant functor

$$F : R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Ab}$$

is *additive* if, for all $f, g : M \to N$,

$$F(f+g) = F(f) + F(g).$$

Equivalently, its maps on morphism sets are group homomorphisms. In particular, $F(0) = 0$ and $F(-f) = -F(f)$. We assume F is additive from now on.

Applying F degree by degree. For a chain complex $C_\bullet$ of left R-modules, write $F(C_\bullet)$ for

$$\cdots \longrightarrow F(C_{n+1}) \xrightarrow{F(d^C_{n+1})} F(C_n) \xrightarrow{F(d^C_n)} F(C_{n-1}) \longrightarrow \cdots .$$

This is a chain complex of abelian groups, since

$$F(d^C_n)F(d^C_{n+1}) = F(d^C_n d^C_{n+1}) = F(0) = 0.$$

Its homology $H_n(F(C_\bullet))$ is an abelian group, equivalently a left $\mathbb{Z}$-module; no additional left R-module structure is assumed. If $f : C_\bullet \to D_\bullet$ is a chain map, then $F(f) = (F(f_n))$ is a chain map $F(C_\bullet) \to F(D_\bullet)$, because

$$F(d^D_n)F(f_n) = F(d^D_n f_n) = F(f_{n-1}d^C_n) = F(f_{n-1})F(d^C_n).$$

Additivity and preservation of composition likewise carry chain homotopies to chain homotopies, and hence preserve chain homotopy equivalences. □

Main motivating example: tensor products. A *right R-module* is an abelian group N with an action $N \times R \to N$, written $(n, r) \mapsto nr$, satisfying

$$(n+n')r = nr + n'r, \quad n(r+r') = nr + nr',$$
$$(nr)r' = n(rr'), \qquad n1_R = n$$

for all $n, n' \in N$ and $r, r' \in R$. A map of right R-modules is *right R-linear* if it is additive and satisfies $f(nr) = f(n)r$. We write **Mod**-R for right R-modules and these maps. Fix a right R-module N for the rest of this example.

Tensor products through universal property. Let M be a left R-module and A an abelian group. A map $\beta : N \times M \to A$ is *biadditive and R-balanced* if

$$\begin{aligned}\beta(n+n',m) &= \beta(n,m)+\beta(n',m),\\ \beta(n,m+m') &= \beta(n,m)+\beta(n,m'),\\ \beta(nr,m) &= \beta(n,rm)\end{aligned}$$

for all $n, n' \in N$, $m, m' \in M$, and $r \in R$. The *tensor product* of N and M over R is an abelian group $N \otimes_R M$ together with a biadditive, R-balanced map

$$\tau : N \times M \longrightarrow N \otimes_R M, \qquad \tau(n,m) = n \otimes m,$$

such that, for every abelian group A, each biadditive, R-balanced map $\beta : N \times M \to A$ factors uniquely through a group homomorphism $\overline{\beta} : N \otimes_R M \to A$:

$$\begin{array}{ccc} N \times M & \xrightarrow{\ \tau\ } & N \otimes_R M \\ & \searrow^{\beta} & \downarrow{\scriptstyle \overline{\beta}} \\ & & A. \end{array}$$

Equivalently, $\overline{\beta}(n \otimes m) = \beta(n,m)$.

Existence and uniqueness. Let L be the free abelian group on symbols $[n,m]$, indexed by $(n,m) \in N \times M$, and let K be the subgroup generated by

$$\begin{aligned}&[n+n',m]-[n,m]-[n',m],\\ &[n,m+m']-[n,m]-[n,m'],\\ &[nr,m]-[n,rm].\end{aligned}$$

Then L/K, with $\tau(n,m) = [n,m]+K$, has the required property: each β extends uniquely to a homomorphism $L \to A$, which vanishes on K and hence factors uniquely through L/K. The elements $n \otimes m$, called *pure tensors*, generate $N \otimes_R M$ as an abelian group.

If (T,τ) and (T',τ') both have the universal property, it gives unique homomorphisms $a : T \to T'$ and $b : T' \to T$ with $a\tau = \tau'$ and $b\tau' = \tau$. Uniqueness forces $ba = \mathrm{id}_T$ and $ab = \mathrm{id}_{T'}$. Thus the tensor product is unique up to a unique isomorphism compatible with its universal map. □

The tensor functor. For an R-linear map $f : M \to M'$, the map $(n,m) \mapsto n \otimes f(m)$ is biadditive and R-balanced, since

$$(nr) \otimes f(m) = n \otimes rf(m) = n \otimes f(rm).$$

The universal property therefore defines a group homomorphism

$$\mathrm{id}_N \otimes f : N \otimes_R M \longrightarrow N \otimes_R M', \qquad n \otimes m \longmapsto n \otimes f(m).$$

On pure tensors one checks

$$\mathrm{id}_N \otimes \mathrm{id}_M = \mathrm{id}_{N \otimes_R M}, \qquad (\mathrm{id}_N \otimes g)(\mathrm{id}_N \otimes f) = \mathrm{id}_N \otimes (g \circ f)$$

for composable f, g. Since pure tensors generate, these are identities of homomorphisms. Thus

$$N \otimes_R - : R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Ab}, \qquad M \longmapsto N \otimes_R M, \quad f \longmapsto \mathrm{id}_N \otimes f$$

is a covariant functor. It is additive: for $f, h : M \to M'$, biadditivity gives

$$\mathrm{id}_N \otimes (f + h) = (\mathrm{id}_N \otimes f) + (\mathrm{id}_N \otimes h). \qquad \square$$

A concrete calculation. Let $m, n \geq 2$, put $B = \mathbb{Z}/m\mathbb{Z}$ and $d = \gcd(m, n)$, and take $F = B \otimes_{\mathbb{Z}} -$. The universal property gives isomorphisms

$$B \otimes_{\mathbb{Z}} \mathbb{Z} \xrightarrow{\cong} B, \qquad x \otimes k \longmapsto kx,$$
$$B \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) \xrightarrow{\cong} B/nB, \quad x \otimes \overline{k} \longmapsto kx + nB.$$

Their inverses send x to $x \otimes 1$ and $x + nB$ to $x \otimes \overline{1}$, respectively; the tensor relations make these maps well defined. Reduction modulo d identifies $B/nB \cong \mathbb{Z}/d\mathbb{Z}$. Apply F to the free resolution

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

Under these identifications, the resulting sequence is

$$0 \longrightarrow B \xrightarrow{\times n} B \xrightarrow{q} B/nB \longrightarrow 0,$$

where q is the quotient map.

First observation: right exactness. The right-hand part remains exact: q is surjective and $\ker q = nB = \mathrm{im}(\times n)$. However, if $d > 1$, multiplication by n kills the nonzero element $\overline{m/d} \in B$, so the left-hand injection is lost.

Definition (Right and left exactness). An additive covariant functor $F : R\text{-}\mathbf{Mod} \to \mathbf{Ab}$ is *right exact* if every exact sequence

$$A \longrightarrow B \longrightarrow C \longrightarrow 0$$

is sent to an exact sequence

$$F(A) \longrightarrow F(B) \longrightarrow F(C) \longrightarrow 0.$$

It is *left exact* if it similarly preserves exactness of sequences $0 \to A \to B \to C$.

Proposition. For every right R-module N, the functor $N \otimes_R -$ is right exact.

Proof. Let $A \xrightarrow{\alpha} B \xrightarrow{\beta} C \to 0$ be an exact sequence of left R-modules. Every pure tensor $x \otimes c$ lifts to $x \otimes b$ with $\beta(b) = c$, so $\mathrm{id}_N \otimes \beta$ is surjective. Put $J = \mathrm{im}(\mathrm{id}_N \otimes \alpha)$. There is an induced homomorphism

$$(N \otimes_R B)/J \longrightarrow N \otimes_R C.$$

Its inverse is induced by the biadditive, R-balanced map

$$N \times C \longrightarrow (N \otimes_R B)/J, \qquad (x, c) \longmapsto [x \otimes b], \quad \beta(b) = c.$$

This is independent of the choice of b: two choices differ by $\alpha(a)$, whose tensor with x belongs to J. Balancing follows from $xr \otimes b = x \otimes rb$ and $\beta(rb) = rc$. The universal property now gives the inverse, so $\ker(\mathrm{id}_N \otimes \beta) = J$, proving exactness. □

Second observation: homology. Return to $F = (\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} -$. Delete $\mathbb{Z}/n\mathbb{Z}$ and its image under F from the two sequences above. The remaining complexes are

$$\begin{aligned} P_\bullet: &\quad 0 \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \longrightarrow 0, \\ F(P_\bullet): &\quad 0 \longrightarrow B \xrightarrow{\times n} B \longrightarrow 0, \end{aligned}$$

with nonzero terms in degrees 1 and 0. As before, $H_1(P_\bullet) = 0$ and $H_0(P_\bullet) \cong \mathbb{Z}/n\mathbb{Z}$. For the transformed complex,

$$\begin{aligned} &H_1(F(P_\bullet)) = \ker(\times n : B \to B) = \langle \overline{m/d} \rangle \cong \mathbb{Z}/d\mathbb{Z}, \\ &H_0(F(P_\bullet)) = B/nB \cong F(\mathbb{Z}/n\mathbb{Z}), \\ &H_i(F(P_\bullet)) = 0 \qquad (i \geq 2). \end{aligned}$$

Here H_1 is exactly the kernel that measures the failure to preserve the original injection. Meanwhile, H_0 simply recovers the functor's value on the resolved left R-module. This suggests the general definition.

Left derived functors. Let $F : R\text{-}\mathbf{Mod} \to \mathbf{Ab}$ be an additive, covariant, right exact functor.

Proposition. For any projective resolution $P_\bullet \xrightarrow{\varepsilon} M$, the sequence

$$F(P_1) \xrightarrow{F(d_1)} F(P_0) \xrightarrow{F(\varepsilon)} F(M) \longrightarrow 0$$

is exact at $F(P_0)$ and $F(M)$, and the augmentation induces an isomorphism $H_0(F(P_\bullet)) \cong F(M)$.

Proof. The sequence $P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \to 0$ is exact, so its image under F is exact by right exactness. Thus $F(\varepsilon)$ is surjective and $\operatorname{im} F(d_1) = \ker F(\varepsilon)$. In the underlying complex $F(P_\bullet)$, the degree-0 differential is zero. The first isomorphism theorem therefore gives

$$H_0(F(P_\bullet)) = F(P_0)/\operatorname{im} F(d_1) = F(P_0)/\ker F(\varepsilon) \cong F(M),$$

where the isomorphism sends $[x]$ to $F(\varepsilon)(x)$. □

Exactness at $F(P_n)$ for $n \geq 1$ can fail. Homology in these degrees measures that failure.

Definition (Values on left modules). For each left R-module M, choose a projective resolution $P_\bullet \xrightarrow{\varepsilon} M$ and define

$$L_nF(M) := H_n(F(P_\bullet)) = \frac{\ker F(d_n)}{\operatorname{im} F(d_{n+1})} \qquad (n \geq 0).$$

Here $P_\bullet$ is the underlying complex, with $d_0 = 0$; the augmentation term $F(M)$ is omitted. The proposition gives $L_0F(M) \cong F(M)$.

Well-definedness on objects. Let $P_\bullet$ and $Q_\bullet$ be two projective resolutions of M. The comparison theorem gives maps $f : P_\bullet \to Q_\bullet$ and $g : Q_\bullet \to P_\bullet$ lifting id_M, with $gf \simeq \mathrm{id}_{P_\bullet}$ and $fg \simeq \mathrm{id}_{Q_\bullet}$. Since F is covariant and additive, it preserves chain homotopies; hence

$$F(g)F(f) \simeq \mathrm{id}_{F(P_\bullet)}, \qquad F(f)F(g) \simeq \mathrm{id}_{F(Q_\bullet)}.$$

It follows that

$$H_n(F(f)) : H_n(F(P_\bullet)) \xrightarrow{\cong} H_n(F(Q_\bullet)).$$

Any two comparison maps lifting id_M are chain homotopic, so this isomorphism is independent of the chosen comparison map. These isomorphisms compose transitively: a composite of comparison maps again lifts id_M. Comparing a resolution with itself gives the identity isomorphism. Thus $L_nF(M) = H_n(F(P_\bullet))$ is independent of the choice of projective resolution $P_\bullet$, up to canonical isomorphism. □

Action on morphisms and functoriality. Given an R-linear map $u : M \to M'$, choose a chain map $f : P_\bullet \to Q_\bullet$ lifting u, where $Q_\bullet$ resolves M'. Define

$$\begin{aligned} L_nF(u) = H_n(F(f)) : \quad & L_nF(M) \longrightarrow L_nF(M'), \\ & [x] \longmapsto [F(f_n)(x)]. \end{aligned}$$

This is independent of the lift: two lifts are chain homotopic, their images under F are chain homotopic, and hence they induce the same homomorphism on homology.

We must also compare different resolutions of both left R-modules. Let $P'_\bullet \to M$ and $Q'_\bullet \to M'$ be projective resolutions, and let $f' : P'_\bullet \to Q'_\bullet$ be a lift of u. Choose comparison maps

$$a : P_\bullet \to P'_\bullet, \qquad b : Q_\bullet \to Q'_\bullet$$

lifting id_M and $\mathrm{id}_{M'}$, respectively. Both bf and $f'a$ are maps $P_\bullet \to Q'_\bullet$ lifting u. Uniqueness in the comparison theorem gives $bf \simeq f'a$. Since F preserves chain homotopies, taking homology gives a commutative square:

$$\begin{array}{ccc} H_n(F(P_\bullet)) & \xrightarrow{H_n(F(f))} & H_n(F(Q_\bullet)) \\ {\scriptstyle H_n(F(a))}\downarrow{\scriptstyle \cong} & & {\scriptstyle \cong}\downarrow{\scriptstyle H_n(F(b))} \\ H_n(F(P'_\bullet)) & \xrightarrow[H_n(F(f'))]{} & H_n(F(Q'_\bullet)). \end{array}$$

The vertical maps are the canonical isomorphisms from the preceding paragraph. Thus the maps obtained using the two pairs of resolutions agree under these identifications. This proves independence of both the lift and the resolutions. □

The identity chain map lifts id_M. If $g : Q_\bullet \to S_\bullet$ lifts $v : M' \to M''$, then $g \circ f$ lifts $v \circ u$. We may therefore use these lifts to compute

$$\begin{aligned} L_nF(\mathrm{id}_M) &= \mathrm{id}_{L_nF(M)}, \\ L_nF(v \circ u) &= H_n(F(g) \circ F(f)) \\ &= L_nF(v) \circ L_nF(u). \end{aligned}$$

For parallel maps u, u', the sum of their lifts lifts $u + u'$; additivity of F and of the induced homology maps gives $L_nF(u + u') = L_nF(u) + L_nF(u')$. Thus, for every $n \geq 0$, we obtain an additive covariant functor

$$L_nF : R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Ab},$$

called the *nth left derived functor* of F. The commuting square above says that different choices of resolutions give functors related by a canonical natural isomorphism. In degree 0, compatibility with the augmentations identifies $L_0F(u)$ with $F(u)$, so $L_0F \cong F$. □

Third observation: tensoring with $\mathbb{Z}$. Return to the standard resolution of $\mathbb{Z}/n\mathbb{Z}$, but now take $N = \mathbb{Z}$ as a right $\mathbb{Z}$-module and put $F = \mathbb{Z} \otimes_{\mathbb{Z}} -$. For every left $\mathbb{Z}$-module M, the tensor relations give an isomorphism

$$\mathbb{Z} \otimes_{\mathbb{Z}} M \xrightarrow{\cong} M, \qquad k \otimes x \longmapsto kx,$$

with inverse $x \mapsto 1 \otimes x$. These identifications commute with homomorphisms. Thus tensoring with $\mathbb{Z}$ preserves exactness, and the standard resolution becomes the same exact sequence under these identifications:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

After deleting the augmentation term, we therefore obtain

$$L_0F(\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}, \qquad L_iF(\mathbb{Z}/n\mathbb{Z}) = 0 \quad (i \geq 1).$$

Here both left and right exactness are preserved. □

Definition (Exact functor). An additive covariant functor $F : R\text{-}\mathbf{Mod} \to \mathbf{Ab}$ is *exact* if it sends every short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

to a short exact sequence

$$0 \longrightarrow F(A) \longrightarrow F(B) \longrightarrow F(C) \longrightarrow 0.$$

Equivalently, F is both left exact and right exact. In particular, $\mathbb{Z} \otimes_{\mathbb{Z}} -$ is exact.

Proposition. An exact functor preserves exactness of sequences of any finite or infinite length.

Proof. Consider any three consecutive terms $A \xrightarrow{a} B \xrightarrow{b} C$ with $\operatorname{im} a = \ker b$. Put $K = \ker b$ and $I = \operatorname{im} b$. Factor $a = \iota q$ and $b = jp$, where $q : A \twoheadrightarrow K$, $j : I \hookrightarrow C$, and

$$0 \longrightarrow K \xrightarrow{\iota} B \xrightarrow{p} I \longrightarrow 0$$

is short exact. Applying F preserves this short exact sequence. The short exact sequences associated with $\ker q \subseteq A$ and $I \subseteq C$ also show that $F(q)$ is surjective and $F(j)$ is injective. Consequently,

$$\operatorname{im} F(a) = \operatorname{im} F(\iota) = \ker F(p) = \ker F(b).$$

This proves exactness at each position, regardless of the sequence's length. □

Proposition (Two vanishing conditions). Let $F : R\text{-}\mathbf{Mod} \to \mathbf{Ab}$ be additive, covariant, and right exact. Then $L_iF(M) = 0$ for every $i \geq 1$ if either

1. F is exact; or
2. M is projective.

In both cases, $L_0F(M) \cong F(M)$.

Proof. If F is exact, it preserves the exactness of any augmented projective resolution of M. Hence its underlying complex $F(P_\bullet)$ is exact in every positive degree, giving $L_iF(M) = 0$ for $i \geq 1$.

If M is projective, choose the length-zero projective resolution

$$0 \longrightarrow M \xrightarrow{\mathrm{id}_M} M \longrightarrow 0,$$

with $P_0 = M$, augmentation id_M, and $P_i = 0$ for $i \geq 1$. After deleting the augmentation term and applying F, the complex has only $F(M)$ in degree 0. Its positive-degree homology is therefore zero, and its degree-0 homology is $F(M)$. □

3 Tor Functors and Group Homology

3.1 Tor functors

Definition. Let N be a right R-module and A a left R-module. For $i \geq 0$, define

$$\operatorname{Tor}_i^R(N, A) := L_i(N \otimes_R -)(A) = H_i(N \otimes_R P_\bullet),$$

where $P_\bullet \xrightarrow{\varepsilon_P} A$ is a projective resolution of left R-modules. The augmentation term $N \otimes_R A$ is omitted before taking homology. By Section 2.3, these abelian groups are independent of the resolution up to canonical isomorphism, and

$$\operatorname{Tor}_0^R(N, A) \cong N \otimes_R A.$$

Tor is covariant in both variables. Functoriality in A is already part of the construction of left derived functors. An additive map $v : N \to N'$ satisfying $v(xr) = v(x)r$, called *right R-linear*, induces the chain map $v \otimes \mathrm{id}_{P_\bullet}$ and hence a map $\operatorname{Tor}_i^R(N, A) \to \operatorname{Tor}_i^R(N', A)$. These maps commute with the maps in the second variable.

Four calculations over $\mathbb{Z}$. Let $n, m \geq 2$ and $d = \gcd(n, m)$. The previous calculations give the following isomorphism types:

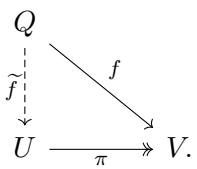
N	A	$\operatorname{Tor}_0^{\mathbb{Z}}(N, A)$	$\operatorname{Tor}_1^{\mathbb{Z}}(N, A)$	$\operatorname{Tor}_i^{\mathbb{Z}}(N, A)$ $(i \geq 2)$
$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	0	0
$\mathbb{Z}$	$\mathbb{Z}/m\mathbb{Z}$	$\mathbb{Z}/m\mathbb{Z}$	0	0
$\mathbb{Z}/n\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}/n\mathbb{Z}$	0	0
$\mathbb{Z}/n\mathbb{Z}$	$\mathbb{Z}/m\mathbb{Z}$	$\mathbb{Z}/d\mathbb{Z}$	$\mathbb{Z}/d\mathbb{Z}$	0

Verification. The first two rows use exactness of $\mathbb{Z} \otimes_{\mathbb{Z}} -$; the third uses projectivity of the second argument $\mathbb{Z}$. For the last row, resolve $\mathbb{Z}/m\mathbb{Z}$ by its standard length-one free resolution. Tensoring gives

$$0 \longrightarrow \mathbb{Z}/n\mathbb{Z} \xrightarrow{\times m} \mathbb{Z}/n\mathbb{Z} \longrightarrow 0$$

in degrees 1 and 0. Both the kernel and the quotient by the image are isomorphic to $\mathbb{Z}/d\mathbb{Z}$, as computed previously. There are no terms in higher degrees. □

Definition (Projective right modules). A right R-module Q is *projective* if, for every surjective right R-linear map $\pi : U \to V$ and every right R-linear map $f : Q \to V$, there is a right R-linear map $\widetilde{f} : Q \to U$ with $\pi\widetilde{f} = f$. The lift need not be unique:



Here Q, U, V are right R-modules and every arrow is right R-linear.

Projective resolutions of right modules. A *projective resolution of right R-modules* is an exact sequence

$$\cdots \longrightarrow Q_2 \longrightarrow Q_1 \longrightarrow Q_0 \xrightarrow{\varepsilon_Q} N \longrightarrow 0$$

of right R-modules and right R-linear maps, with every Q_i projective. As before, $Q_\bullet$ denotes the complex with N deleted. Free right R-modules have the form $\bigoplus_{s\in S} e_s R$. The same basis-lifting proof makes them projective right R-modules, and the same successive free-cover construction gives a projective resolution of every right R-module.

Lemma (Tensoring with projective left or right modules). If N is a projective right R-module, then $N \otimes_R -$ is exact on left R-modules. If P is a projective left R-module, then $- \otimes_R P$ is exact on right R-modules.

Proof. We prove the statement for $N \otimes_R -$. Section 2.3 already proves right exactness, so it remains to show that this functor preserves monomorphisms.

Let N be projective and let $i : A \to B$ be a monomorphism of left R-modules. We use two facts about $E = \mathbb{Q}/\mathbb{Z}$ proved in Appendix A: additive maps into E extend across inclusions of abelian groups, and such maps separate points of every abelian group. For a left R-module X, let $X^\vee$ be the abelian group of additive maps $X \to E$. Give it the right R-action

$$(\chi r)(x) = \chi(rx).$$

The restriction map

$$i^\vee : B^\vee \longrightarrow A^\vee, \qquad \chi \longmapsto \chi \circ i$$

is right R-linear and surjective by the extension property of E.

Given $0 \neq z \in N \otimes_R A$, choose an additive map $\lambda : N \otimes_R A \to E$ with $\lambda(z) \neq 0$. The formula

$$\phi : N \longrightarrow A^\vee, \qquad \phi(x)(a) = \lambda(x \otimes a)$$

defines a right R-linear map, since $\phi(xr)(a) = \lambda(x \otimes ra) = (\phi(x)r)(a)$. By the defining lifting property of N, there is a right R-linear map $\psi : N \to B^\vee$ with $i^\vee\psi = \phi$. The map $(x, b) \mapsto \psi(x)(b)$ is biadditive and R-balanced:

$$\psi(xr)(b) = (\psi(x)r)(b) = \psi(x)(rb).$$

The tensor universal property therefore gives an additive map

$$\widetilde{\lambda} : N \otimes_R B \longrightarrow E, \qquad \widetilde{\lambda}(x \otimes b) = \psi(x)(b).$$

On pure tensors,

$$\widetilde{\lambda}(x \otimes i(a)) = (i^\vee\psi(x))(a) = \phi(x)(a) = \lambda(x \otimes a).$$

Thus $\widetilde{\lambda} \circ (\mathrm{id}_N \otimes i) = \lambda$. Since $\lambda(z) \neq 0$, we have $(\mathrm{id}_N \otimes i)(z) \neq 0$. This proves injectivity and hence exactness. The proof for a projective left R-module P is analogous, with the sides reversed, and is omitted. □

Theorem (Balanced computation of Tor). Let N be a right R-module and A a left R-module. Let $P_\bullet \xrightarrow{\varepsilon_P} A$ be a projective resolution of left R-modules and $Q_\bullet \xrightarrow{\varepsilon_Q} N$ a projective resolution of right R-modules. For every $i \geq 0$, there is a canonical isomorphism

$$\mathrm{Tor}_i^R(N, A) = H_i(N \otimes_R P_\bullet) \cong H_i(Q_\bullet \otimes_R A),$$

compatible with maps in both N and A. Thus either variable may be resolved, with its indicated module side.

Proof. Form the *total complex* $T_\bullet$ of the array $(Q_p \otimes_R P_q)_{p,q\geq 0}$ by setting

$$T_k = \bigoplus_{p+q=k} Q_p \otimes_R P_q,$$
$$\partial(x \otimes y) = d_Q x \otimes y + (-1)^p x \otimes d_P y \qquad (x \in Q_p,\ y \in P_q).$$

Here the underlying differentials in degree 0 are zero. The sign makes the two mixed terms in ∂^2 cancel, so $\partial^2 = 0$. The augmentations give chain maps, surjective in every degree,

$$N \otimes_R P_\bullet \xleftarrow{\alpha} T_\bullet \xrightarrow{\beta} Q_\bullet \otimes_R A.$$

The map α is $\varepsilon_Q \otimes \mathrm{id}$ on summands with $p = 0$ and zero elsewhere; β is $\mathrm{id} \otimes \varepsilon_P$ on summands with $q = 0$ and zero elsewhere.

We first show that $\ker \alpha$ has zero homology. The kernel is taken degreewise and forms a complex because α is a chain map. For fixed q, tensoring $Q_\bullet \to N$ with the projective left R-module P_q preserves exactness. In $\ker \alpha$, the $p = 0$ term of this row is replaced by $\ker(\varepsilon_Q \otimes \mathrm{id}_{P_q})$; all other terms remain unchanged. Each resulting row is exact. For a cycle z of total degree k, remove its components successively in the order

$$(0, k),\ (1, k-1),\ \ldots,\ (k, 0).$$

At $(p, k-p)$, the cycle equation makes the component horizontally closed, since the preceding components have been removed. Row exactness supplies a horizontal preimage in $(p+1, k-p)$. Subtracting its total boundary removes the chosen component and changes only the next component, when present. After $k+1$ steps nothing remains, so z was a boundary.

Likewise, tensoring $P_\bullet \to A$ with each projective right R-module Q_p makes the columns of $\ker \beta$ exact. This time remove components in the order $(k, 0), (k-1, 1), \ldots, (0, k)$, using vertical preimages multiplied by $(-1)^p$ to match the sign in ∂. Thus $\ker \beta$ also has zero homology.

For completeness, a chain map $\rho : C_\bullet \to D_\bullet$ that is surjective in each degree and whose kernel has zero homology induces an isomorphism on homology. Indeed, lift a cycle $y \in D_k$ to $x \in C_k$. Then dx is a cycle in $\ker \rho$, so $dx = dw$ for some $w \in (\ker \rho)_k$; hence $x - w$ is a cycle lifting y. Conversely, if a cycle x satisfies $\rho(x) = dy$, lift y to t. Then $x - dt$ is a cycle in $\ker \rho$, hence a boundary there, so x is a boundary in $C_\bullet$. Applying this observation to α and β gives

$$H_i(N \otimes_R P_\bullet) \xleftarrow{\cong} H_i(T_\bullet) \xrightarrow{\cong} H_i(Q_\bullet \otimes_R A).$$

Finally, lift left R-linear maps in A and right R-linear maps in N to their resolutions by the comparison theorem, whose proof applies on either side. Tensoring the lifts gives a map between the total complexes commuting with α and β. The resulting isomorphism therefore respects maps in both variables. Taking identity maps on N and A compares different resolutions. Homotopic comparison lifts induce the same maps on homology, so the isomorphism is independent of these choices. □

3.2 Group homology

The group ring. Fix a group G. Its *integral group ring* is

$$R = \mathbb{Z}G = \left\{ \sum_{g \in G} n_g g : n_g \in \mathbb{Z}, \quad n_g = 0 \text{ for all but finitely many } g \right\}.$$

Addition is coefficientwise, and multiplication is defined by

$$\left(\sum_g n_g g \right) \left(\sum_h m_h h \right) = \sum_{g,h} n_g m_h (gh).$$

This is an associative unital ring, with identity $1 \cdot 1_G$. Giving a left R-module A is equivalent to giving an abelian group with a G-action by additive automorphisms: restrict the left R-action to G, or extend the group action by

$$\left(\sum_g n_g g \right) a = \sum_g n_g (ga).$$

An additive map between such left R-modules is R-linear exactly when it commutes with the G-action. We study G through its actions on A: G is the object of study, and A is the coefficient left R-module.

Trivial left and right R-modules and the augmentation ideal. Give $\mathbb{Z}$ the trivial left and right actions $g \cdot k = k \cdot g = k$ for all $g \in G$ and $k \in \mathbb{Z}$. Define the *augmentation*

$$\varepsilon : R \longrightarrow \mathbb{Z}, \qquad \varepsilon \left(\sum_g n_g g \right) = \sum_g n_g = \left(\sum_g n_g g \right) \cdot 1 = 1 \cdot \left(\sum_g n_g g \right).$$

It is a surjective unital ring homomorphism: $\varepsilon(rs) = \varepsilon(r)\varepsilon(s)$ and $\varepsilon(1_G) = 1$.
Its kernel is the two-sided *augmentation ideal*

$$\begin{aligned} I := \ker \varepsilon &= \left\{ \sum_g n_g g \in R : \sum_g n_g = 0 \right\} \\ &= \langle g - 1 : g \in G \rangle_{\mathbb{Z}}. \end{aligned}$$

Indeed, every $g - 1$ lies in I, and an element with coefficient sum zero satisfies $\sum_g n_g g = \sum_g n_g (g - 1)$. The induced ring isomorphism is

$$R/I \xrightarrow{\cong} \mathbb{Z}, \qquad r + I \longmapsto \varepsilon(r),$$

with inverse $k \mapsto k1_G + I$, since $r - \varepsilon(r)1_G \in I$. The trivial left and right R-actions on $\mathbb{Z}$ are

$$r \cdot k = \varepsilon(r)k, \qquad k \cdot r = k\varepsilon(r).$$

Thus this same identification $R/I \cong \mathbb{Z}$ is an isomorphism of left R-modules and of right R-modules. □

If $G \neq \{1\}$, the abelian group $\mathbb{Z}$ with trivial G-actions is not free as either a left or a right $R = \mathbb{Z}G$-module.

Proof. Choose $g \in G$ with $g \neq 1$. Since distinct group elements are distinct basis elements of the free abelian group $\mathbb{Z}G$, we have $g - 1 \neq 0$ in R. For either trivial R-action, however, g acts as the identity, so

$$(g-1)k = 0, \qquad k(g-1) = 0 \qquad \text{for every } k \in \mathbb{Z}.$$

Suppose that $\mathbb{Z}$ were a free left R-module with basis $(e_s)_{s\in S}$. Since $\mathbb{Z} \neq 0$, the set S is nonempty. Choose $s \in S$. In a free left R-module, an element is zero exactly when all its basis coefficients are zero. The element $(g-1)e_s$ has the nonzero coefficient $g-1$ at e_s, so $(g-1)e_s \neq 0$. This contradicts the fact that $g-1$ annihilates every element of the trivial left R-module.

For a free right R-module, the same argument applies to $e_s(g-1)$: its coefficient at e_s is again $g-1 \neq 0$, whereas the trivial right action makes it zero. Thus neither the trivial left nor the trivial right R-module is free. □

For $G = \{1\}$, we have $R = \mathbb{Z}$, and $\mathbb{Z}$ is free of rank one as both a left and a right R-module.

Coinvariants. To force the action on A to be trivial, impose the relations $ga = a$. The resulting *coinvariants* are

$$A_G := A/IA, \qquad IA = \langle ga - a : g \in G,\ a \in A\rangle_{\mathbb{Z}}.$$

Here IA consists of finite sums of elements ia, with $i \in I$ and $a \in A$; it is an R-submodule since I is a two-sided ideal. Write $\overline{a} = a + IA$. Then $\overline{ga} = \overline{a}$, so the induced action on A_G is trivial.

Proposition (Coinvariants as a tensor product). There are inverse isomorphisms

$$\phi_A : A_G \longrightarrow \mathbb{Z} \otimes_R A, \qquad \overline{a} \longmapsto 1 \otimes a,$$
$$\psi_A : \mathbb{Z} \otimes_R A \longrightarrow A_G, \qquad k \otimes a \longmapsto k\overline{a},$$

where $\mathbb{Z}$ is the trivial *right* R-module.

Proof. The map ϕ_A is well defined because the additive map $a \mapsto 1 \otimes a$ kills each generator of IA:

$$1 \otimes (ga - a) = (1 \cdot g) \otimes a - 1 \otimes a = 0.$$

For ψ_A, the map $(k, a) \mapsto k\overline{a}$ is biadditive and R-balanced. Indeed, $\overline{ra} = \varepsilon(r)\overline{a}$, so

$$(k \cdot r)\overline{a} = k\varepsilon(r)\overline{a} = k\overline{ra}.$$

The tensor universal property therefore gives a well-defined ψ_A. Finally,

$$\psi_A\phi_A(\overline{a}) = \overline{a}, \qquad \phi_A\psi_A(k \otimes a) = 1 \otimes ka = k \otimes a.$$

Since pure tensors generate, the maps are inverse. □

The balancing relation transfers the action on A to the left tensor factor $\mathbb{Z}$, where the right G-action is trivial.

The coinvariant functor. For an R-linear map $f : A \to B$, we have $f(IA) \subseteq IB$, so

$$f_G : A_G \longrightarrow B_G, \qquad f_G(\overline{a}) = \overline{f(a)}$$

is well defined. Induced quotient maps preserve identities, composition, and addition. Thus

$$(-)_G : R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Ab}, \qquad A \longmapsto A_G, \quad f \longmapsto f_G$$

is additive and covariant. □

A canonical natural isomorphism. The isomorphisms ϕ_A form a *natural isomorphism of functors*

$$\phi : (-)_G \xrightarrow{\cong} \mathbb{Z} \otimes_R -.$$

This means that each component ϕ_A is an isomorphism and that, for every R-linear map $f : A \to B$, the following square commutes:

$$\begin{array}{ccc} A_G & \xrightarrow[\cong]{\phi_A} & \mathbb{Z} \otimes_R A \\ {\scriptstyle f_G}\downarrow & & \downarrow{\scriptstyle \mathrm{id}_{\mathbb{Z}} \otimes f} \\ B_G & \xrightarrow[\phi_B]{\cong} & \mathbb{Z} \otimes_R B. \end{array}$$

Both paths send $\overline{a}$ to $1 \otimes f(a)$. The isomorphism is *canonical* because its formula $\phi_A(\overline{a}) = 1 \otimes a$ requires no choice of basis or resolution. After identifying the values on objects by these maps, the actions of the two functors on morphisms agree as well.

This identification transfers right exactness. If $A \to B \to C \to 0$ is exact, tensoring gives an exact sequence $\mathbb{Z} \otimes_R A \to \mathbb{Z} \otimes_R B \to \mathbb{Z} \otimes_R C \to 0$. The naturality squares identify its terms and arrows with those of $A_G \to B_G \to C_G \to 0$, which is therefore exact. Thus $(-)_G$ is right exact. □

Definition (Group homology). For $n \geq 0$, the *nth homology of G with coefficients in A* is

$$H_n(G; A) := L_n((-)_G)(A).$$

Let $P_\bullet \to A$ be a projective resolution of left R-modules, and omit the augmentation term. Naturality applied to each differential $d_n : P_n \to P_{n-1}$ gives

$$\phi_{P_{n-1}} \circ (d_n)_G = (\mathrm{id}_{\mathbb{Z}} \otimes d_n) \circ \phi_{P_n}.$$

Hence the component maps form an *isomorphism of chain complexes*

$$\phi_{P_\bullet} : (P_\bullet)_G \xrightarrow{\cong} \mathbb{Z} \otimes_R P_\bullet.$$

Taking homology gives

$$\begin{aligned} H_n(G; A) &= H_n((P_\bullet)_G) \\ &\cong H_n(\mathbb{Z} \otimes_R P_\bullet) = \mathrm{Tor}_n^{\mathbb{Z}G}(\mathbb{Z}, A). \end{aligned}$$

Here $\mathbb{Z}$ in the tensor product is the trivial right R-module. In particular,

$$H_0(G; A) \cong A_G.$$

For fixed G, each $H_n(G; -)$ is an additive covariant functor from left R-modules to abelian groups.

The identification with Tor also respects coefficient maps. For $u : A \to B$, choose a projective resolution $Q_\bullet \to B$ and a chain lift $f : P_\bullet \to Q_\bullet$. Naturality applied to each f_n gives

$$\phi_{Q_\bullet} \circ f_G = (\mathrm{id}_{\mathbb{Z}} \otimes f) \circ \phi_{P_\bullet}.$$

Taking homology therefore identifies $H_n(G;u)$ with $\operatorname{Tor}_n^{\mathbb{Z}G}(\mathbb{Z},u)$ under the displayed isomorphisms. Thus the tensor description computes both the objects and the maps of group homology. □

Computing with a resolution of $\mathbb{Z}$. By the balanced computation theorem, any projective resolution $Q_\bullet \to \mathbb{Z}$ of the trivial right R-module gives

$$H_n(G;A) \cong H_n(Q_\bullet \otimes_R A).$$

Thus we can fix one resolution for G and use it for every coefficient left R-module A. In Chapter 5 we will construct the *bar resolution*, an explicit free resolution of this trivial right R-module.

Proposition (Vanishing in positive degrees). For $n > 0$, the following hold.

1. If $G = \{1\}$, then $H_n(G;A) = 0$ for every abelian group A.
2. If A is projective over $\mathbb{Z}G$ (in particular, if it is free over $\mathbb{Z}G$), then $H_n(G;A) = 0$.

Proof. In the first case, $R = \mathbb{Z}G = \mathbb{Z}$, so the trivial right R-module $\mathbb{Z}$ is free of rank one, hence projective. Tensoring with a projective right R-module is exact. Thus $\mathbb{Z} \otimes_R -$ is exact, and the canonical natural isomorphism $(-)_G \cong \mathbb{Z} \otimes_R -$ shows that $(-)_G$ is exact as well. Its higher left derived functors therefore vanish.

In the second case, use the projective resolution $0 \to A \xrightarrow{\mathrm{id}_A} A \to 0$. Its underlying complex, and hence its coinvariant complex, is concentrated in degree 0, so its positive-degree homology vanishes. □

Example: the infinite cyclic group with trivial coefficients. Let $G = \langle t \rangle \cong \mathbb{Z}$ and $R = \mathbb{Z}G$, the ring of finite Laurent polynomials $\sum_k a_k t^k$ with $a_k \in \mathbb{Z}$. Take $A = \mathbb{Z}$ with the trivial left R-action. We compute directly from a free resolution of the coefficient left R-module A.

A free resolution of the coefficient left R-module. Consider the sequence of left R-modules

$$0 \longrightarrow R \xrightarrow{\times(t-1)} R \xrightarrow{\varepsilon} A \longrightarrow 0, \qquad \varepsilon(f) = f(1).$$

The augmentation is surjective and has kernel $(t-1)R$. Indeed, if $f = \sum_k a_k t^k$ has $f(1) = 0$, then $f = \sum_k a_k(t^k - 1)$. Each $t^k - 1$ is a multiple of $t-1$, also for negative k, since $t^{-j} - 1 = -t^{-j}(t^j - 1)$ for $j > 0$. Conversely, every multiple of $t - 1$ evaluates to zero at 1. Multiplication by $t - 1$ is injective: if $a_m t^m$ is the highest nonzero term of f, then $(t-1)f$ has the nonzero term $a_m t^{m+1}$. Thus the sequence is exact, and its two copies of R are free left R-modules of rank one. Write $P_1 = P_0 = R$ and $P_n = 0$ for $n \geq 2$. □

Apply coinvariants. Omit the augmentation term. Each nonzero term has coinvariants

$$(P_i)_G = R_G = R/(\ker\varepsilon)R = R/(\ker\varepsilon) = R/(t-1) \cong \mathbb{Z} \qquad (i = 0, 1),$$

and the induced differential is zero:

$$(d_1)_G(\overline{r}) = \overline{(t-1)r} = 0.$$

The resulting chain complex is therefore

$$0 \longrightarrow \underbrace{\mathbb{Z}}_{\text{degree } 1} \xrightarrow{0} \underbrace{\mathbb{Z}}_{\text{degree } 0} \longrightarrow 0.$$

By the definition of group homology,

$$H_n(G;\mathbb{Z}) = H_n((P_\bullet)_G) \cong \begin{cases} \mathbb{Z}, & n = 0, 1, \\ 0, & n \geq 2. \end{cases}$$

In degree zero this agrees with $A_G = \mathbb{Z}$. The original differential d_1 is injective, but $(d_1)_G$ is zero: coinvariants destroy exactness at degree one, and the nonzero $H_1(G;\mathbb{Z})$ records this failure. □

Warning (Projectivity depends on the ring). Although $A = \mathbb{Z}$ is free as an abelian group, it is not projective as a left R-module. If it were projective, the surjection $\varepsilon : R \to A$ would admit an R-linear section σ. Then

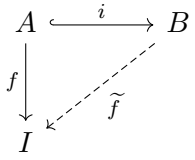
$$(t-1)\sigma(1) = \sigma((t-1)\cdot 1) = 0.$$

Injectivity of multiplication by $t-1$ gives $\sigma(1) = 0$, contradicting $\varepsilon\sigma(1) = 1$. In particular, the abelian-group decomposition $R = \mathbb{Z}\cdot 1 \oplus I$ is not a decomposition of left R-modules: $\mathbb{Z}\cdot 1$ is not stable under multiplication by t. A free resolution over $\mathbb{Z}$ cannot simply be used as a projective resolution over $\mathbb{Z}G$. □

4 From Injective Resolutions to Group Cohomology

4.1 Injective left modules and injective resolutions

Definition by the extension property. A left R-module I is *injective* if, for every monomorphism $i : A \to B$ of left R-modules and every R-linear map $f : A \to I$, there exists an R-linear map $\widetilde{f} : B \to I$ such that $\widetilde{f} i = f$, that is, the following diagram commutes:



The extension need not be unique. This property is dual to projectivity.

Cochain complexes and cohomology. A *cochain complex* of left R-modules is a sequence

$$C^\bullet = (\, \cdots \longrightarrow C^{n-1} \xrightarrow{d_C^{n-1}} C^n \xrightarrow{d_C^n} C^{n+1} \longrightarrow \cdots), \qquad d_C^{n+1} d_C^n = 0.$$

Its degree-n *cohomology* is the left R-module

$$H^n(C^\bullet) = \ker d_C^n / \operatorname{im} d_C^{n-1}.$$

Elements of $\ker d_C^n$ are *n-cocycles*; elements of $\operatorname{im} d_C^{n-1}$ are *n-coboundaries*. Thus $H^n(C^\bullet) = 0$ precisely when the complex is exact at C^n. For a cochain complex of abelian groups, its cohomology is an abelian group.

Definition and notation. An *injective resolution* of a left R-module M, written $M \xrightarrow{\iota} I^\bullet$, is an exact sequence

$$0 \longrightarrow M \xrightarrow{\iota} I^0 \xrightarrow{d_I^0} I^1 \xrightarrow{d_I^1} I^2 \longrightarrow \cdots$$

of left R-modules in which every I^n is injective. The map ι is the *coaugmentation*. The notation $I^\bullet$ alone denotes the underlying cochain complex

$$0 \longrightarrow I^0 \xrightarrow{d_I^0} I^1 \xrightarrow{d_I^1} I^2 \longrightarrow \cdots,$$

with $I^n = 0$ for $n < 0$. It excludes M. The coaugmentation identifies M with $H^0(I^\bullet)$, and $H^n(I^\bullet) = 0$ for $n > 0$. In the full coaugmented complex, M occupies degree -1 and $d_I^{-1} = \iota$; all lower terms are zero.

Existence. The following **fact** is proved in Appendix A: **every left R-module embeds into an injective left R-module.** Thus R-**Mod** has *enough injectives*. Embed M into I^0, then embed $I^0/\iota(M)$ into an injective left R-module I^1. Their composite defines $d_I^0 : I^0 \to I^1$, with $\ker d_I^0 = \iota(M)$. Repeating this construction with the successive cokernels gives an injective resolution. □

Remark. The enough-injectives theorem takes more work than the existence of free left R-modules, and explicit injective resolutions can be difficult to construct. For Ext and group cohomology, the computation theorem in Section 4.3 will let us use a projective, usually free,

resolution in the other argument. This does not replace injective resolutions for an arbitrary right derived functor.

Cochain maps. A *cochain map* $f : C^\bullet \to D^\bullet$ is a family of R-linear maps $f^n : C^n \to D^n$ satisfying

$$d_D^n f^n = f^{n+1} d_C^n \qquad \text{for every } n.$$

Equivalently, the squares below commute:

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & C^{n-1} & \xrightarrow{d_C^{n-1}} & C^n & \xrightarrow{d_C^n} & C^{n+1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle f^{n-1}} & & \downarrow{\scriptstyle f^n} & & \downarrow{\scriptstyle f^{n+1}} & & \\
\cdots & \longrightarrow & D^{n-1} & \xrightarrow[d_D^{n-1}]{} & D^n & \xrightarrow[d_D^n]{} & D^{n+1} & \longrightarrow & \cdots
\end{array}$$

Proposition. A cochain map induces an R-linear map

$$H^n(f) : H^n(C^\bullet) \longrightarrow H^n(D^\bullet), \qquad [x] \longmapsto [f^n(x)].$$

Moreover, $H^n(gf) = H^n(g)H^n(f)$ and $H^n(\mathrm{id}_{C^\bullet}) = \mathrm{id}_{H^n(C^\bullet)}$.

Proof. The cochain map identities send cocycles to cocycles and coboundaries to coboundaries. Hence the formula is well-defined; composition and identity maps are preserved by the same formula. □

Cochain homotopies. Let $f, g : C^\bullet \to D^\bullet$ be cochain maps. A *cochain homotopy* from f to g is a family of R-linear maps $s^n : C^n \to D^{n-1}$ such that

$$f^n - g^n = d_D^{n-1} s^n + s^{n+1} d_C^n \qquad \text{for every } n.$$

We write $f \simeq g$. For complexes concentrated in nonnegative degrees, $s^0 : C^0 \to D^{-1} = 0$ is zero. In the diagram, the middle vertical arrow is the sum of the two composites using a dashed arrow:

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & C^{n-1} & \xrightarrow{d_C^{n-1}} & C^n & \xrightarrow{d_C^n} & C^{n+1} & \longrightarrow & \cdots \\
 & & & \swarrow^{s^n} & \downarrow{\scriptstyle f^n - g^n} & \swarrow^{s^{n+1}} & & & \\
\cdots & \longrightarrow & D^{n-1} & \xrightarrow[d_D^{n-1}]{} & D^n & \xrightarrow[d_D^n]{} & D^{n+1} & \longrightarrow & \cdots
\end{array}$$

Proposition. Cochain homotopic maps induce the same maps on cohomology.

Proof. For $x \in \ker d_C^n$, we have $f^n(x) - g^n(x) = d_D^{n-1} s^n(x) \in \operatorname{im} d_D^{n-1}$.

Thus $H^n(f)[x] = H^n(g)[x]$. □

Cochain homotopy equivalences. A cochain map $f : C^\bullet \to D^\bullet$ is a *cochain homotopy equivalence* if there is a cochain map $g : D^\bullet \to C^\bullet$ such that

$$gf \simeq \mathrm{id}_{C^\bullet}, \qquad fg \simeq \mathrm{id}_{D^\bullet}.$$

In this case we write $C^\bullet \simeq D^\bullet$. The map g is a *homotopy inverse* of f.

Proposition. A cochain homotopy equivalence induces isomorphisms on cohomology.

Proof. If g is a homotopy inverse of f, then

$$H^n(g)H^n(f) = H^n(gf) = H^n(\mathrm{id}_{C^\bullet}) = \mathrm{id}_{H^n(C^\bullet)},$$
$$H^n(f)H^n(g) = H^n(fg) = H^n(\mathrm{id}_{D^\bullet}) = \mathrm{id}_{H^n(D^\bullet)}.$$

The middle equalities use the two homotopies. Hence $H^n(g)$ is the inverse of $H^n(f)$. □

Theorem (Comparison of injective resolutions). Let $M \xrightarrow{\iota_I} I^\bullet$ and $N \xrightarrow{\iota_J} J^\bullet$ be injective resolutions of left R-modules. For every R-linear map $u : M \to N$, the following hold.

1. There is a cochain map $f : I^\bullet \to J^\bullet$ such that $f^0\iota_I = \iota_J u$. Together they define a cochain map
$$\widetilde{u} = (u, f) : (M \xrightarrow{\iota_I} I^\bullet) \longrightarrow (N \xrightarrow{\iota_J} J^\bullet),$$
with component u in degree -1 and f^n in degree $n \geq 0$. We call it an *extension of* u.

2. Any two such extensions (u, f) and (u, f') are cochain homotopic. The homotopy can be chosen with $\widetilde{s}^0 : I^0 \to N$ equal to zero; its remaining components $s^n : I^n \to J^{n-1}$, $n \geq 1$, give a homotopy $f \simeq f'$ after the coaugmentation terms are deleted.

Proof (Existence). through the

Proof (Existence). All maps in this proof are homomorphisms of left R-modules. Start with the two coaugmented resolutions and the given map u:

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & M & \xrightarrow{\iota_I} & I^0 & \xrightarrow{d_I^0} & I^1 & \xrightarrow{d_I^1} & I^2 & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle u} & & & & & & & & \\
0 & \longrightarrow & N & \xrightarrow[\iota_J]{} & J^0 & \xrightarrow[d_J^0]{} & J^1 & \xrightarrow[d_J^1]{} & J^2 & \longrightarrow & \cdots
\end{array}$$

We seek maps $f^n : I^n \to J^n$ extending u to a cochain map of these coaugmented complexes. First, $\iota_J u : M \to J^0$ is R-linear and $\iota_I : M \hookrightarrow I^0$ is a monomorphism. Injectivity of J^0 gives an extension $f^0 : I^0 \to J^0$ making the following diagram commute:

$$\begin{array}{ccc}
M & \xrightarrow{\iota_I} & I^0 \\
{\scriptstyle \iota_J u}\downarrow & \swarrow_{f^0} & \\
J^0 & &
\end{array}$$

Thus $f^0\iota_I = \iota_J u$, which is the cochain map condition at the coaugmentation.

Next, consider $d_J^0 f^0 : I^0 \to J^1$. We check that

$$d_J^0 f^0 \iota_I = d_J^0 \iota_J u = 0.$$

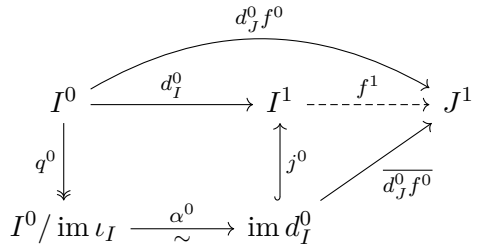
Thus this map vanishes on $\operatorname{im}\iota_I = \ker d_I^0$ and factors through the quotient $I^0/\operatorname{im}\iota_I$. Write $q^0 : I^0 \twoheadrightarrow I^0/\operatorname{im}\iota_I$ for the quotient map, and let

$$\alpha^0 : I^0/\operatorname{im}\iota_I \xrightarrow{\sim} \operatorname{im} d_I^0, \qquad [x] \longmapsto d_I^0(x).$$

Exactness makes α^0 an isomorphism of left R-modules. If $j^0 : \operatorname{im} d_I^0 \hookrightarrow I^1$ is the inclusion, then $d_I^0 = j^0 \alpha^0 q^0$. The factorization of $d_J^0 f^0$, transported along α^0, is

$$\overline{d_J^0 f^0} : \operatorname{im} d_I^0 \longrightarrow J^1, \qquad d_I^0(x) \longmapsto d_J^0 f^0(x).$$

Here the bar changes the domain, by passing to a quotient and identifying that quotient with $\operatorname{im} d_I^0$; the codomain remains J^1. This is dual to the projective proof, where bars changed the codomain by restricting it to the appropriate image or kernel. Injectivity of J^1 extends this map across j^0, giving the dashed arrow:



By construction, $f^1 j^0 = \overline{d_J^0 f^0}$. Therefore

$$f^1 d_I^0 = f^1 j^0 \alpha^0 q^0 = \overline{d_J^0 f^0} \alpha^0 q^0 = d_J^0 f^0,$$

which is the cochain map condition in degree 0. At each subsequent step, the identities already constructed give

$$d_J^n f^n d_I^{n-1} = d_J^n d_J^{n-1} f^{n-1} = 0 \qquad (n \geq 1).$$

Since $\ker d_I^n = \operatorname{im} d_I^{n-1}$, the map $d_J^n f^n$ factors through $I^n/\operatorname{im} d_I^{n-1} \cong \operatorname{im} d_I^n$ and extends across $\operatorname{im} d_I^n \hookrightarrow I^{n+1}$ by injectivity of J^{n+1}. Repeating this construction gives all f^n, with $f^{n+1} d_I^n = d_J^n f^n$ for $n \geq 0$. Thus $f = (f^n)$ is a cochain map, and $f^0 \iota_I = \iota_J u$ makes $\widetilde{u} = (u, f)$ the required coaugmented extension. The formal induction is omitted.

(Uniqueness up to homotopy). Let $\widetilde{u} = (u, f)$ and $\widetilde{u}' = (u, f')$ be two extensions. On the underlying complexes, set $s^0 : I^0 \to J^{-1} = 0$ to zero.

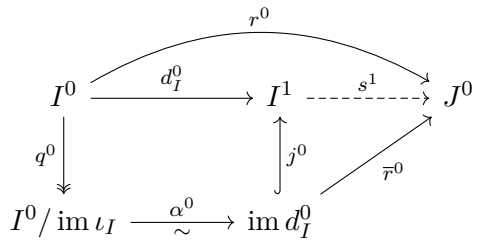
Degree 0. Put $r^0 = f^0 - f'^0$. Since both maps extend u,

$$r^0 \iota_I = \iota_J u - \iota_J u = 0.$$

Hence r^0 factors through q^0 and, using the same α^0, defines the map

$$\overline{r}^0 : \operatorname{im} d_I^0 \longrightarrow J^0, \qquad d_I^0(x) \longmapsto r^0(x).$$

As in the existence proof, the bar changes the domain through the quotient identification. Injectivity of J^0 gives the dashed extension $s^1 : I^1 \to J^0$:



Thus $s^1 j^0 = \overline{r}^0$, and composing with $\alpha^0 q^0$ gives

$$f^0 - f'^0 = r^0 = \overline{r}^0 \alpha^0 q^0 = s^1 d_I^0.$$

Degree 1. Put $r^1 = f^1 - f'^1 - d_J^0 s^1$. The cochain map identities and the degree-0 identity give

$$r^1 d_I^0 = d_J^0 (f^0 - f'^0 - s^1 d_I^0) = 0.$$

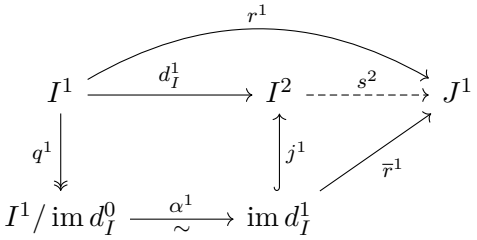
Exactness therefore gives the factorization through the quotient map $q^1 : I^1 \twoheadrightarrow I^1/\operatorname{im} d_I^0$ and the isomorphism

$$\alpha^1 : I^1/\operatorname{im} d_I^0 \xrightarrow{\sim} \operatorname{im} d_I^1, \qquad [x] \longmapsto d_I^1(x).$$

Write $j^1 : \operatorname{im} d_I^1 \hookrightarrow I^2$ for the inclusion. Then $d_I^1 = j^1 \alpha^1 q^1$, and the induced map is

$$\overline{r}^1 : \operatorname{im} d_I^1 \longrightarrow J^1, \qquad d_I^1(x) \longmapsto r^1(x).$$

Injectivity of J^1 extends it across j^1 to $s^2 : I^2 \to J^1$:



Consequently,

$$r^1 = \overline{r}^1 \alpha^1 q^1 = s^2 d_I^1, \qquad f^1 - f'^1 = d_J^0 s^1 + s^2 d_I^1.$$

Degree 2 *and continuation.* Put $r^2 = f^2 - f'^2 - d_J^1 s^2$. The additional term still vanishes:

$$r^2 d_I^1 = d_J^1 (f^1 - f'^1 - s^2 d_I^1) = d_J^1 d_J^0 s^1 = 0.$$

The same factorization and extension produce $s^3 : I^3 \to J^2$. Continuing gives $f^n - f'^n = d_J^{n-1} s^n + s^{n+1} d_I^n$ for all $n \geq 0$. Again, the formal induction is omitted. On the coaugmented complexes, choose $\widetilde{s}^0 : I^0 \to N$ and all lower homotopy components to be zero, and put $\widetilde{s}^n = s^n$ for $n \geq 1$. The degree--1 identity is $u - u = \widetilde{s}^0 \iota_I = 0$, and the additional degree-0 term is $\iota_J \widetilde{s}^0 = 0$. Thus this homotopy works both before and after deleting M, N. □

Corollary (Uniqueness of injective resolutions). Any two injective resolutions of a left R-module M are cochain homotopy equivalent, both as coaugmented complexes and as underlying complexes.

Proof. The comparison theorem extends id_M in both directions to cochain maps $f : I^\bullet \to J^\bullet$ and $g : J^\bullet \to I^\bullet$. The composites gf and fg extend id_M, as do $\mathrm{id}_{I^\bullet}$ and $\mathrm{id}_{J^\bullet}$, respectively. Uniqueness up to homotopy gives

$$gf \simeq \mathrm{id}_{I^\bullet}, \qquad fg \simeq \mathrm{id}_{J^\bullet}.$$

The theorem supplies homotopies with the maps from degree 0 to the coaugmentation term equal to zero, so the same equivalence holds with or without the copies of M. □

Remark. As for projective resolutions, the useful conclusion is homotopy equivalence. An additive covariant functor preserves its defining identities, even when it does not preserve exactness.

4.2 Right derived functors

Let $F : R\text{-}\mathbf{Mod} \to \mathbf{Ab}$ be additive, covariant, and left exact. Thus it sends every short exact sequence $0 \to A \to B \to C \to 0$ to a sequence

$$0 \longrightarrow F(A) \longrightarrow F(B) \longrightarrow F(C)$$

that is exact at $F(A)$ and $F(B)$; the last map need not be surjective.

Applying F to cochain complexes. Applying F degree by degree sends a cochain complex $C^\bullet$ to a cochain complex $F(C^\bullet)$ of abelian groups:

$$F(d^{n+1})F(d^n) = F(d^{n+1}d^n) = F(0) = 0.$$

It sends cochain maps to cochain maps and cochain homotopies to cochain homotopies. Indeed, applying F to the homotopy identity gives

$$F(f^n) - F(g^n) = F(d_D^{n-1})F(s^n) + F(s^{n+1})F(d_C^n).$$

Consequently, F preserves cochain homotopy equivalences. The cohomology $H^n(F(C^\bullet))$ is an abelian group; we do not assume an additional left R-module structure. □

Proposition (Degree zero). For an injective resolution $M \xrightarrow{\iota} I^\bullet$, the sequence

$$0 \longrightarrow F(M) \xrightarrow{F(\iota)} F(I^0) \xrightarrow{F(d^0)} F(I^1)$$

is exact at $F(M)$ and $F(I^0)$. In particular, $F(\iota)$ induces an isomorphism $F(M) \cong H^0(F(I^\bullet))$.

Proof. The sequence $0 \to M \xrightarrow{\iota} I^0 \xrightarrow{d^0} I^1$ is exact. Left exactness gives that $F(\iota)$ is injective and

$$\operatorname{im} F(\iota) = \ker F(d^0) = H^0(F(I^\bullet)),$$

where the last equality uses the underlying complex, with no term in degree -1. The isomorphism sends x to $F(\iota)(x)$. □

Exactness at $F(I^n)$ for $n \geq 1$ may fail. Its failure is measured by cohomology in these degrees.

Definition (Values on left modules). For each left R-module M, choose an injective resolution $M \xrightarrow{\iota} I^\bullet$ and define

$$R^nF(M) := H^n(F(I^\bullet)) = \frac{\ker F(d^n)}{\operatorname{im} F(d^{n-1})} \qquad (n \geq 0).$$

Here $I^\bullet$ is the underlying complex, with $d^{-1} = 0$; the coaugmentation term $F(M)$ is omitted before taking cohomology. The preceding proposition gives $R^0F(M) \cong F(M)$.

Well-definedness on objects. Two injective resolutions $I^\bullet, J^\bullet$ of M admit comparison maps $f : I^\bullet \to J^\bullet$ and $g : J^\bullet \to I^\bullet$ extending id_M, with $gf \simeq \mathrm{id}_{I^\bullet}$ and $fg \simeq \mathrm{id}_{J^\bullet}$. Since F preserves these homotopies,

$$H^n(F(f)) : H^n(F(I^\bullet)) \xrightarrow{\cong} H^n(F(J^\bullet))$$

has inverse $H^n(F(g))$. Any two comparison maps extending id_M are homotopic, so the isomorphism is independent of that choice. Comparison maps compose to comparison maps, so these

identifications also agree when a third resolution is introduced. Comparing a resolution with itself gives the identity isomorphism. Thus $R^nF(M)$ is independent of the injective resolution, up to canonical isomorphism. $\square$

Action on morphisms and functoriality. Given an R-linear map $u : M \to N$, choose injective resolutions $M \xrightarrow{\iota_I} I^\bullet$ and $N \xrightarrow{\iota_J} J^\bullet$, and a cochain map $f : I^\bullet \to J^\bullet$ extending u. Define

$$\begin{aligned} R^nF(u) := H^n(F(f)) : \quad & R^nF(M) \longrightarrow R^nF(N), \\ & [x] \longmapsto [F(f^n)(x)]. \end{aligned}$$

This is independent of the extension: two extensions are cochain homotopic, their images under F are cochain homotopic, and hence they induce the same homomorphism on cohomology.

We must also compare different resolutions of both left R-modules. Let $M \to I'^\bullet$ and $N \to J'^\bullet$ be injective resolutions, and let $f' : I'^\bullet \to J'^\bullet$ extend u. Choose comparison maps

$$a : I^\bullet \to I'^\bullet, \qquad b : J^\bullet \to J'^\bullet$$

extending id_M and id_N, respectively. Both bf and $f'a$ are maps $I^\bullet \to J'^\bullet$ extending u. Uniqueness in the comparison theorem gives $bf \simeq f'a$. Since F preserves cochain homotopies, taking cohomology gives a commutative square:

$$\begin{array}{ccc} H^n(F(I^\bullet)) & \xrightarrow{H^n(F(f))} & H^n(F(J^\bullet)) \\ {\scriptstyle H^n(F(a))}\downarrow{\scriptstyle \cong} & & {\scriptstyle \cong}\downarrow{\scriptstyle H^n(F(b))} \\ H^n(F(I'^\bullet)) & \xrightarrow[H^n(F(f'))]{} & H^n(F(J'^\bullet)). \end{array}$$

The vertical maps are the canonical isomorphisms from the preceding paragraph. Thus the maps obtained using the two pairs of resolutions agree under these identifications. This proves independence of both the extension and the resolutions. $\square$

The identity cochain map extends id_M. If $g : J^\bullet \to K^\bullet$ extends $v : N \to T$, where $K^\bullet$ resolves T, then $g \circ f$ extends $v \circ u$. We may therefore use these extensions to compute

$$\begin{aligned} R^nF(\mathrm{id}_M) &= \mathrm{id}_{R^nF(M)}, \\ R^nF(v \circ u) &= H^n(F(g) \circ F(f)) \\ &= R^nF(v) \circ R^nF(u). \end{aligned}$$

For parallel maps $u, u' : M \to N$, the sum of their extensions between the same resolutions extends $u + u'$. Additivity of F and of the induced cohomology maps gives $R^nF(u + u') = R^nF(u) + R^nF(u')$. Thus, for every $n \geq 0$, we obtain an additive covariant functor

$$R^nF : R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Ab},$$

called the *nth right derived functor* of F. The commuting square above says that different choices of resolutions give functors related by a canonical natural isomorphism.

In degree zero, let $\alpha_M : F(M) \xrightarrow{\cong} R^0F(M)$ be the isomorphism induced by $F(\iota_I)$. The extension identity $f^0\iota_I = \iota_J u$ gives

$$F(f^0)F(\iota_I) = F(\iota_J)F(u), \qquad R^0F(u)\alpha_M = \alpha_N F(u).$$

Thus the degree-zero identifications are natural, so $R^0F \cong F$. □

Proposition (Two vanishing conditions). For every $n > 0$, we have $R^nF(M) = 0$ if either

1. F is exact; or
2. M is injective.

In both cases, $R^0F(M) \cong F(M)$.

Proof. An exact functor preserves exact sequences of arbitrary length, as proved in Section 2.3. Thus, in the first case, $F(I^\bullet)$ remains exact in positive degrees. In the second case, choose the length-zero injective resolution

$$0 \longrightarrow M \xrightarrow{\mathrm{id}_M} M \longrightarrow 0,$$

where the second copy is I^0 and $I^n = 0$ for $n > 0$. After omitting the first copy and applying F, only $F(M)$ in degree zero remains. □

4.3 Hom and Ext functors

The Hom functors. For left R-modules A and B, let

$$\mathrm{Hom}_R(A, B) = \{\phi : A \to B : \phi \text{ is } R\text{-linear}\}.$$

Pointwise addition makes this an abelian group. For a general noncommutative ring R, it need not carry the pointwise left R-module structure. Fixing A gives an additive covariant functor

$$\mathrm{Hom}_R(A, -) : R\textbf{-Mod} \longrightarrow \textbf{Ab}, \qquad v : B \to B' \longmapsto (v_* : \phi \mapsto v\phi).$$

Fixing B instead gives an additive *contravariant* functor: an arrow $u : A \to A'$ induces an arrow in the opposite direction,

$$u^* : \mathrm{Hom}_R(A', B) \longrightarrow \mathrm{Hom}_R(A, B), \qquad \phi \longmapsto \phi u.$$

It preserves identities and reverses the order of composition.

Proposition (Left exactness of Hom). For every short exact sequence of left R-modules $0 \to X \xrightarrow{i} Y \xrightarrow{p} Z \to 0$, the sequences

$$0 \longrightarrow \mathrm{Hom}_R(A, X) \xrightarrow{i_*} \mathrm{Hom}_R(A, Y) \xrightarrow{p_*} \mathrm{Hom}_R(A, Z),$$
$$0 \longrightarrow \mathrm{Hom}_R(Z, B) \xrightarrow{p^*} \mathrm{Hom}_R(Y, B) \xrightarrow{i^*} \mathrm{Hom}_R(X, B)$$

are exact. This is the meaning of left exactness for the two Hom functors, with arrows reversed in the contravariant case. If A is projective, the first sequence is also surjective at the last term. If B is injective, the second sequence is also surjective there. Thus $\mathrm{Hom}_R(A, -)$ is exact when A is projective, and $\mathrm{Hom}_R(-, B)$ is exact when B is injective.

Proof. The induced homomorphisms of abelian groups are

$$i_*(f) = i \circ f, \qquad p_*(g) = p \circ g, \qquad p^*(h) = h \circ p, \qquad i^*(g) = g \circ i.$$

If $i_*(f) = 0$, injectivity of i gives $f = 0$, so i_* is injective. Since $pi = 0$, we have $\operatorname{im} i_* \subseteq \ker p_*$. Conversely, let $g : A \to Y$ satisfy $pg = 0$. For every $a \in A$, there is a unique $x \in X$ with $i(x) = g(a)$, because $g(a) \in \ker p = \operatorname{im} i$. Define $f(a) = x$. Additivity and R-linearity of f follow from those of g and injectivity of i. Thus $g = if$, proving $\ker p_* = \operatorname{im} i_*$.

If $p^*(h) = 0$, surjectivity of p gives $h = 0$, so p^* is injective. Again $pi = 0$ gives $\operatorname{im} p^* \subseteq \ker i^*$. Conversely, let $g : Y \to B$ satisfy $gi = 0$. Define

$$h : Z \longrightarrow B, \qquad h(z) = g(y), \quad p(y) = z.$$

Two choices of y differ by an element of $\ker p = \operatorname{im} i$, on which g vanishes. Thus h is well defined. Choosing lifts of sums and scalar multiples shows that it is additive and R-linear. Hence $g = hp$, proving $\ker i^* = \operatorname{im} p^*$.

If the left R-module A is projective, each map $A \to Z$ lifts through p, so p_* is surjective. If the left R-module B is injective, each map $X \to B$ extends across i, so i^* is surjective. These are exactly the defining lifting and extension properties. □

Definition (Ext). For $n \geq 0$, define

$$\operatorname{Ext}^n_R(A, B) := R^n \operatorname{Hom}_R(A, -)(B) = H^n\big(\operatorname{Hom}_R(A, I^\bullet)\big),$$

where $B \xrightarrow{\iota_I} I^\bullet$ is an injective resolution. The coaugmentation term $\operatorname{Hom}_R(A, B)$ is omitted before taking cohomology. Section 4.2 gives independence of this choice up to canonical isomorphism and

$$\operatorname{Ext}^0_R(A, B) \cong \operatorname{Hom}_R(A, B).$$

Ext is contravariant in A and covariant in B. Explicitly, $u : A' \to A$ acts by $\phi \mapsto \phi \circ u$ on $\operatorname{Hom}_R(A, I^\bullet)$. For $v : B \to B'$, choose a cochain map $g : I^\bullet \to J^\bullet$ extending v; in degree n, the induced map is $\phi \mapsto g^n \circ \phi$. Hence there are commuting induced maps

$$u^* : \operatorname{Ext}^n_R(A, B) \longrightarrow \operatorname{Ext}^n_R(A', B), \qquad v_* : \operatorname{Ext}^n_R(A, B) \longrightarrow \operatorname{Ext}^n_R(A, B').$$

Proposition (Vanishing). For $n > 0$, we have $\operatorname{Ext}^n_R(A, B) = 0$ if either A is projective or B is injective.

Proof. If A is projective, the functor $\operatorname{Hom}_R(A, -)$ is exact. If B is injective, use its length-zero injective resolution. In either case the vanishing conditions of Section 4.2 apply. □

Theorem (Equivalent computation of Ext). Let A and B be left R-modules, and let $P_\bullet \xrightarrow{\varepsilon_P} A$ be a projective resolution of left R-modules. Then, for every $n \geq 0$, there is a canonical isomorphism

$$\operatorname{Ext}^n_R(A, B) \cong H^n\big(\operatorname{Hom}_R(P_\bullet, B)\big),$$

compatible with maps in both variables. Here the cochain complex on the right is

$$0 \longrightarrow \operatorname{Hom}_R(P_0, B) \xrightarrow{\delta^0} \operatorname{Hom}_R(P_1, B) \xrightarrow{\delta^1} \operatorname{Hom}_R(P_2, B) \longrightarrow \cdots,$$
$$\delta^n(\phi) = \phi d^P_{n+1}.$$

The term $\operatorname{Hom}_R(A, B)$ is omitted. Thus one may resolve the second variable injectively or the first variable projectively; both resolutions consist of left R-modules.

Proof. Choose an injective resolution $B \xrightarrow{\iota_I} I^\bullet$. For $p, q \geq 0$, put $C^{p,q} = \mathrm{Hom}_R(P_p, I^q)$ and form the total cochain complex

$$T^n = \bigoplus_{p+q=n} C^{p,q},$$
$$D\phi = h\phi + (-1)^p v\phi, \qquad h\phi = \phi d^P_{p+1}, \quad v\phi = d^q_I \phi \quad (\phi \in C^{p,q}).$$

The two operations h and v commute and square to zero; the sign therefore gives $D^2 = 0$. The augmentation of $P_\bullet$ and coaugmentation of $I^\bullet$ induce cochain maps, injective in every degree,

$$\mathrm{Hom}_R(A, I^\bullet) \xrightarrow{\alpha} T^\bullet \xleftarrow{\beta} \mathrm{Hom}_R(P_\bullet, B).$$

The map α sends ϕ to $\phi\varepsilon_P$ in the $p = 0$ summand; β sends ψ to $\iota_I\psi$ in the $q = 0$ summand. All their other components are zero.

We show that the two quotient complexes have zero cohomology. Since I^q is injective, applying $\mathrm{Hom}_R(-, I^q)$ to the augmented resolution of A gives an exact row

$$0 \longrightarrow \mathrm{Hom}_R(A, I^q) \longrightarrow C^{0,q} \xrightarrow{h} C^{1,q} \xrightarrow{h} C^{2,q} \longrightarrow \cdots .$$

Consequently the rows of $T^\bullet / \operatorname{im}\alpha$ are exact: their first terms are $C^{0,q}/\operatorname{im}\mathrm{Hom}_R(A, I^q)$. For a cocycle of total degree n, remove its components in the order

$$(n, 0),\ (n-1, 1),\ \ldots,\ (1, n-1).$$

At $(p, n-p)$ the cocycle equation makes the component horizontally closed, because components with larger p have already been removed. Row exactness supplies a horizontal preimage in $(p-1, n-p)$. Subtracting its total coboundary removes the chosen component and changes only the next component. The final component at $(0, n)$ is horizontally closed and hence zero by exactness at the start of its row. This also handles $n = 0$. Every cocycle is therefore a coboundary.

Likewise, projectivity of each P_p makes $\mathrm{Hom}_R(P_p, -)$ exact, so the columns of $T^\bullet / \operatorname{im}\beta$ are exact. Remove components in the order $(0, n), (1, n-1), \ldots, (n-1, 1)$, using vertical preimages with the factor $(-1)^p$ to match D. Exactness at the start of the last column forces the remaining component at $(n, 0)$ to vanish. Thus this quotient also has zero cohomology.

An injective cochain map whose quotient has zero cohomology induces an isomorphism on cohomology. Indeed, identify its source with a subcomplex $K^\bullet \subseteq T^\bullet$. A cocycle in T^n becomes a coboundary in $T^\bullet/K^\bullet$; lifting a preimage and subtracting its coboundary gives a cocycle in K^n. Conversely, if a cocycle in K^n equals Dy in T^n, then the image of y is a cocycle in the quotient. Subtracting a lifted coboundary from y makes it lie in K^{n-1} without changing Dy. In degree 0, there are no coboundaries to check. Applying this observation to α and β gives

$$H^n\big(\mathrm{Hom}_R(A, I^\bullet)\big) \xrightarrow{\cong} H^n(T^\bullet) \xleftarrow{\cong} H^n\big(\mathrm{Hom}_R(P_\bullet, B)\big).$$

Finally, lift $u : A' \to A$ to a chain map $f : P'_\bullet \to P_\bullet$ and extend $v : B \to B'$ to a cochain map $g : I^\bullet \to J^\bullet$. The maps $\phi \mapsto g\phi f$ on the double complexes commute with α and β, proving compatibility with both variables. Taking u and v to be identities compares choices of

resolutions. Homotopic comparison maps induce identical maps on cohomology: for example, if $f - f' = ds + sd$, the maps $\phi \mapsto \phi \circ s_{n-1}$ give a cochain homotopy between f^* and $(f')^*$ in degree n. Thus the isomorphism is independent of all comparison choices. □

In particular, computing Ext requires no explicit injective resolution: we may choose a convenient free resolution of its first argument.

Four calculations over $\mathbb{Z}$. Let $n, m \geq 2$ and $d = \gcd(n, m)$. Then

A	B	$\mathrm{Ext}^0_{\mathbb{Z}}(A,B)$	$\mathrm{Ext}^1_{\mathbb{Z}}(A,B)$	$\mathrm{Ext}^i_{\mathbb{Z}}(A,B)$ $(i \geq 2)$
$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	0	0
$\mathbb{Z}$	$\mathbb{Z}/m\mathbb{Z}$	$\mathbb{Z}/m\mathbb{Z}$	0	0
$\mathbb{Z}/n\mathbb{Z}$	$\mathbb{Z}$	0	$\mathbb{Z}/n\mathbb{Z}$	0
$\mathbb{Z}/n\mathbb{Z}$	$\mathbb{Z}/m\mathbb{Z}$	$\mathbb{Z}/d\mathbb{Z}$	$\mathbb{Z}/d\mathbb{Z}$	0

Verification. For the first two rows, $\mathbb{Z}$ is projective and evaluation at 1 identifies $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, B)$ with B. For the other rows, apply $\mathrm{Hom}_{\mathbb{Z}}(-, B)$ to the standard free resolution of $\mathbb{Z}/n\mathbb{Z}$ and omit the augmentation term. The resulting cochain complex is

$$0 \longrightarrow B \xrightarrow{\times n} B \longrightarrow 0$$

in degrees 0 and 1. Its cohomology is $\{b \in B : nb = 0\}$ in degree 0, B/nB in degree 1, and zero in higher degrees. Substituting $B = \mathbb{Z}$ and $B = \mathbb{Z}/m\mathbb{Z}$ gives the table. □

4.4 Group cohomology

Fix a group G and put $R = \mathbb{Z}G$. As in Section 3.2, coefficients are left R-modules, or equivalently abelian groups with a G-action. Write $\varepsilon : R \to \mathbb{Z}$ for the augmentation and $I = \ker \varepsilon$ for its ideal. Here $\mathbb{Z}$ carries the trivial *left* R-module structure, so $R/I \cong \mathbb{Z}$ as left R-modules.

Invariants. The *invariants* of a left R-module A are

$$A^G := \{a \in A : ga = a \text{ for every } g \in G\} = \{a \in A : Ia = 0\}.$$

The equality follows because I is generated by the elements $g - 1$. This is the largest submodule of A on which G acts trivially. Thus invariants retain the elements already fixed by the action, whereas coinvariants impose relations making the action trivial.

Proposition (Invariants as Hom). There are inverse isomorphisms of abelian groups

$$\begin{aligned} \theta_A : A^G &\longrightarrow \mathrm{Hom}_R(\mathbb{Z}, A), \quad a \longmapsto (k \mapsto ka), \\ \eta_A : \mathrm{Hom}_R(\mathbb{Z}, A) &\longrightarrow A^G, \qquad\quad h \longmapsto h(1). \end{aligned}$$

Proof. If $a \in A^G$, then $ra = \varepsilon(r)a$ for every $r \in R$. Hence $k \mapsto ka$ is R-linear, since

$$\theta_A(a)(r \cdot k) = \varepsilon(r)ka = r(ka) = r((\theta_A(a))(k)).$$

Conversely, an R-linear $h : \mathbb{Z} \to A$ satisfies $gh(1) = h(g \cdot 1) = h(1)$, so $h(1) \in A^G$. Finally, $h(k) = kh(1)$ proves that the maps are inverse. They are additive. □

The invariant functor. For an R-linear map $u : A \to B$, restriction gives

$$u^G : A^G \longrightarrow B^G, \qquad a \longmapsto u(a),$$

because $gu(a) = u(ga) = u(a)$ for $a \in A^G$. Restrictions preserve identities, composition, and addition. Thus

$$(-)^G : R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Ab}, \qquad A \longmapsto A^G, \quad u \longmapsto u^G$$

is additive and covariant. □

A canonical natural isomorphism. The isomorphisms θ_A form a *natural isomorphism of functors*

$$\theta : (-)^G \xrightarrow{\cong} \mathrm{Hom}_R(\mathbb{Z}, -).$$

Here $\mathbb{Z}$ is the trivial *left* R-module. For every R-linear map $u : A \to B$, the following square commutes:

$$\begin{array}{ccc} A^G & \xrightarrow[\cong]{\theta_A} & \mathrm{Hom}_R(\mathbb{Z}, A) \\ {\scriptstyle u^G}\downarrow & & \downarrow{\scriptstyle \mathrm{Hom}_R(\mathbb{Z},u)} \\ B^G & \xrightarrow[\theta_B]{\cong} & \mathrm{Hom}_R(\mathbb{Z}, B). \end{array}$$

Both paths send a to the map $k \mapsto ku(a)$. The isomorphism is *canonical* because its formula $\theta_A(a)(k) = ka$ requires no choice of basis or resolution. After identifying the values on objects by these maps, the actions of the two functors on morphisms agree as well.

This identification transfers left exactness. If $0 \to A \to B \to C$ is exact, applying $\mathrm{Hom}_R(\mathbb{Z}, -)$ gives an exact sequence

$$0 \longrightarrow \mathrm{Hom}_R(\mathbb{Z}, A) \longrightarrow \mathrm{Hom}_R(\mathbb{Z}, B) \longrightarrow \mathrm{Hom}_R(\mathbb{Z}, C).$$

The naturality squares identify its terms and arrows with those of $0 \to A^G \to B^G \to C^G$, which is therefore exact at A^G and B^G. Thus $(-)^G$ is left exact. □

Definition (Group cohomology). For $n \geq 0$, the *nth cohomology of G with coefficients in A* is

$$H^n(G; A) := R^n\big((-)^G\big)(A).$$

Let $A \to I^\bullet$ be an injective resolution of left R-modules, and omit the coaugmentation term. Naturality applied to each differential $d^n : I^n \to I^{n+1}$ gives

$$\theta_{I^{n+1}} \circ (d^n)^G = \mathrm{Hom}_R(\mathbb{Z}, d^n) \circ \theta_{I^n}.$$

Hence the component maps form an *isomorphism of cochain complexes*

$$\theta_{I^\bullet} : (I^\bullet)^G \xrightarrow{\cong} \mathrm{Hom}_R(\mathbb{Z}, I^\bullet).$$

Taking cohomology gives

$$\begin{aligned} H^n(G; A) &= H^n\big((I^\bullet)^G\big) \\ &\cong H^n\big(\mathrm{Hom}_R(\mathbb{Z}, I^\bullet)\big) = \mathrm{Ext}^n_{\mathbb{Z}G}(\mathbb{Z}, A). \end{aligned}$$

In particular,

$$H^0(G; A) \cong A^G.$$

For fixed G, each $H^n(G; -)$ is an additive covariant functor from left R-modules to abelian groups.

The identification with Ext also respects coefficient maps. For $u : A \to B$, choose an injective resolution $B \to K^\bullet$ and a cochain map $f : I^\bullet \to K^\bullet$ extending u. Naturality applied to each f^n gives

$$\theta_{K^\bullet} \circ f^G = \mathrm{Hom}_R(\mathbb{Z}, f) \circ \theta_{I^\bullet}.$$

Taking cohomology therefore identifies $H^n(G; u)$ with $\mathrm{Ext}^n_{\mathbb{Z}G}(\mathbb{Z}, u)$ under the displayed isomorphisms. Thus the Hom description computes both the objects and the maps of group cohomology. □

Computing with a resolution of $\mathbb{Z}$. By the equivalent computation theorem for Ext, any projective resolution $P_\bullet \to \mathbb{Z}$ of the trivial *left* R-module gives

$$H^n(G; A) \cong H^n(\mathrm{Hom}_R(P_\bullet, A)).$$

The augmentation term is omitted, and the cochain differential is $\delta^n(h) = h \circ d_{n+1}$. We can therefore fix one resolution for G and use it for every coefficient left R-module A. The computation theorem is natural in A: a coefficient map $u : A \to B$ is computed by the cochain map

$$\mathrm{Hom}_R(P_\bullet, u) : \mathrm{Hom}_R(P_\bullet, A) \longrightarrow \mathrm{Hom}_R(P_\bullet, B), \qquad h \longmapsto u \circ h.$$

In Chapter 5 we will construct the *left bar resolution* for this purpose, alongside the right-module version used for homology.

Proposition (Vanishing in positive degrees). For $n > 0$, the following hold.

1. If $G = \{1\}$, then $H^n(G; A) = 0$ for every abelian group A.
2. If A is injective over $\mathbb{Z}G$, then $H^n(G; A) = 0$.

Proof. In the first case, $R = \mathbb{Z}G = \mathbb{Z}$, so the trivial left R-module $\mathbb{Z}$ is free of rank one, hence projective. Therefore $\mathrm{Hom}_R(\mathbb{Z}, -)$ is exact. The canonical natural isomorphism $(-)^G \cong \mathrm{Hom}_R(\mathbb{Z}, -)$ shows that $(-)^G$ is exact as well. Its higher right derived functors therefore vanish.

In the second case, use the injective resolution $0 \to A \xrightarrow{\mathrm{id}_A} A \to 0$. After omitting the coaugmentation term, the underlying complex and its invariant complex are concentrated in degree 0, so the positive-degree cohomology vanishes. □

Example: the infinite cyclic group with trivial rational coefficients. Let $G = \langle t \rangle \cong \mathbb{Z}$, $R = \mathbb{Z}G$, and $A = \mathbb{Q}$ with the trivial left R-action. We compute directly from an injective resolution of the coefficient left R-module A.

An auxiliary fact: $\mathbb{Q}$ is injective over $\mathbb{Z}$. This fact will produce the injective left R-modules used in the resolution. It does not assert that the trivial left R-module $\mathbb{Q}$ is injective.

Proof. Let $B \subseteq C$ be abelian groups and $h : B \to \mathbb{Q}$ a homomorphism. Order its extensions $h_D : D \to \mathbb{Q}$, with $B \subseteq D \subseteq C$, by extension. A chain has an upper bound obtained by taking the union, so Zorn's lemma gives a maximal extension. If $D \neq C$, choose $x \in C \setminus D$ and write

$$\{k \in \mathbb{Z} : kx \in D\} = m\mathbb{Z}, \qquad m \geq 0.$$

If $m = 0$, choose $q = 0$; if $m > 0$, choose $q = h_D(mx)/m$. Define

$$h' : D + \mathbb{Z}x \longrightarrow \mathbb{Q}, \qquad h'(d + kx) = h_D(d) + kq.$$

This is well defined: if $d+kx = d'+k'x$, then $(k-k')x = d'-d \in D$ and $h_D((k-k')x) = (k-k')q$ by the choice of q. Thus h' extends h_D to a strictly larger subgroup, contradicting maximality. Hence $D = C$, proving the extension property. □

An injective left R-module of sequences. Let

$$I = \{f : \mathbb{Z} \to \mathbb{Q}\}, \qquad (t \cdot f)(k) = f(k + 1),$$

with pointwise addition. Thus I consists of all two-sided rational sequences, and t acts by shifting the index. The shift is an automorphism, so this defines a left R-module.

To prove that I is injective, let $B \subseteq C$ be left R-modules and $\phi : B \to I$ an R-linear map. Evaluation at 0 gives an abelian-group homomorphism $h : B \to \mathbb{Q}$, $h(b) = \phi(b)(0)$. By the preceding fact, extend it to $\widetilde{h} : C \to \mathbb{Q}$ and define

$$\widetilde{\phi} : C \longrightarrow I, \qquad \widetilde{\phi}(c)(k) = \widetilde{h}(t^k c).$$

This map is additive and commutes with t, since

$$\widetilde{\phi}(tc)(k) = \widetilde{h}(t^{k+1}c) = (t \cdot \widetilde{\phi}(c))(k).$$

Hence it is R-linear. For $b \in B$,

$$\widetilde{\phi}(b)(k) = h(t^k b) = \phi(t^k b)(0) = \phi(b)(k),$$

so it extends ϕ. Therefore I is injective over R. □

An injective resolution of the coefficient left R-module. Consider

$$0 \longrightarrow \mathbb{Q} \xrightarrow{\iota} I \xrightarrow{\Delta} I \longrightarrow 0,$$

where

$$\iota(q)(k) = q, \qquad (\Delta f)(k) = f(k + 1) - f(k).$$

Both maps commute with the shift and are R-linear. The map ι is injective, and $\ker \Delta$ consists exactly of constant sequences, so $\ker \Delta = \operatorname{im} \iota$. Given any $b \in I$, set $f(0) = 0$ and solve $f(k + 1) = f(k) + b(k)$ recursively in both directions. This defines a rational value at every integer and gives $\Delta f = b$. Thus Δ is surjective and the sequence is exact. Since both copies of I are injective, this is an injective resolution with $I^0 = I^1 = I$ and $I^n = 0$ for $n \geq 2$. □

Apply invariants. Omit the coaugmentation term $\mathbb{Q}$. The invariant sequences are precisely the constant sequences:

$$I^G = \{f : f(k + 1) = f(k) \text{ for all } k\} \cong \mathbb{Q}.$$

The restriction Δ^G is zero, so the invariant cochain complex is

$$0 \longrightarrow \underbrace{\mathbb{Q}}_{\text{degree } 0} \xrightarrow{0} \underbrace{\mathbb{Q}}_{\text{degree } 1} \longrightarrow 0.$$

Directly from the definition,

$$H^n(G;\mathbb{Q}) = H^n\big((I^\bullet)^G\big) \cong \begin{cases} \mathbb{Q}, & n = 0, 1, \\ 0, & n \geq 2. \end{cases}$$

The degree-zero result agrees with $A^G = \mathbb{Q}$. The nonzero degree-one cohomology records the failure of invariants to preserve the surjection Δ: the constant sequence 1 has the preimage $f(k) = k$, but no invariant preimage. □

Remark (Injectivity depends on the ring). The coefficient left R-module need not be injective; the terms of its injective resolution must be. Here $\mathbb{Q}$ is injective over $\mathbb{Z}$, and I is injective over $\mathbb{Z}G$. The trivial left $\mathbb{Z}G$-module $\mathbb{Q}$ is not injective, since $H^1(G;\mathbb{Q}) \cong \mathbb{Q} \neq 0$. □

5 The Bar Resolution

5.1 The bar resolution

Fix a group G and put $R = \mathbb{Z}G$. The *bar resolution* is an explicit free resolution of the trivial left or right R-module $\mathbb{Z}$, constructed from tuples in G. We first give the resolution by left R-modules, then the one by right R-modules.

The left bar modules. For $n \geq 0$, let

$$B_n^L = \bigoplus_{(g_1,\ldots,g_n)\in G^n} R[g_1|\cdots|g_n].$$

These are free left R-modules; the brackets are formal basis symbols indexed by ordered tuples, with repetitions and identity entries allowed. In degree zero there is one empty bracket, so $B_0^L = R[\,] \cong R$. When both sides occur, we write the left basis symbols as $[g_1|\cdots|g_n]_L$. We construct the differentials by the process in Section 1.3: map a free left R-module onto $\mathbb{Z}$, then seek generators of each successive kernel.

Degree one: generators of the augmentation kernel. Give $\mathbb{Z}$ its trivial left R-action. The first surjection is

$$\varepsilon : B_0^L \longrightarrow \mathbb{Z}, \qquad \varepsilon(r[\,]) = \varepsilon_R(r),$$

where $\varepsilon_R : R \to \mathbb{Z}$ is the coefficient-sum map. Its kernel is $I[\,]$, where $I = \ker \varepsilon_R$ is generated as a left ideal by the elements $g - 1$. Thus a free left R-module mapping onto this kernel can have one basis symbol $[g]$ for each $g \in G$, sent to $(g-1)[\,]$. This explains the choice of B_1^L and the R-linear map

$$d_1^L : B_1^L \longrightarrow B_0^L, \qquad d_1^L[g] = (g-1)[\,].$$

In particular, $\operatorname{im} d_1^L = \ker \varepsilon$.

Degree two: relations among these generators. The next map must have image $\ker d_1^L$. We therefore look for formal combinations of the symbols $[g]$ with zero image in B_0^L. For $g, h \in G$, splitting the difference $gh - 1$ at g gives

$$gh - 1 = (g-1) + (gh - g) = (g-1) + g(h-1).$$

Consequently $[gh]$ and $[g] + g[h]$ have the same image under d_1^L. Their difference

$$r(g,h) := g[h] - [gh] + [g]$$

is an element of $\ker d_1^L$. Here $g[h]$ means scalar multiplication of the basis symbol $[h]$, while $[gh]$ is the basis symbol indexed by the product gh. The relation is represented by the element $r(g,h)$ of the free left R-module B_1^L; it is its image under d_1^L that is zero.

Introduce one basis symbol $[g|h]$ for each such relation and define

$$d_2^L : B_2^L \longrightarrow B_1^L, \qquad d_2^L[g|h] = r(g,h) = g[h] - [gh] + [g].$$

Then $d_1^L d_2^L = 0$. We still need to prove that these relations generate the whole kernel; the exactness argument below will do this in every degree.

Degree three: relations among relations. Now we seek elements of $\ker d_2^L$. The definition of $r(g,h)$ gives the actual equality

$$[gh] = [g] + g[h] - r(g,h) \qquad \text{in } B_1^L.$$

For three elements g, h, k, apply this identity twice to $[ghk]$. We can first split $(gh)k$ and then gh, or first split $g(hk)$ and then hk. Keeping the correction terms gives

$$\begin{aligned}
[ghk] &= [gh] + gh[k] - r(gh,k) \\
&= [g] + g[h] + gh[k] - r(g,h) - r(gh,k), \\
[ghk] &= [g] + g[hk] - r(g,hk) \\
&= [g] + g[h] + gh[k] - g\,r(h,k) - r(g,hk).
\end{aligned}$$

The common expression $[g] + g[h] + gh[k]$ shows that the total correction terms agree:

$$r(g,h) + r(gh,k) = g\,r(h,k) + r(g,hk).$$

Replacing each r by the corresponding degree-two basis symbol therefore gives an element of $\ker d_2^L$. We record it with a new symbol $[g|h|k]$ and set

$$d_3^L[g|h|k] = g[h|k] - [gh|k] + [g|hk] - [g|h].$$

Applying d_2^L gives zero by the displayed identity. Thus each new bar records a relation among the preceding generators. □

Vertex coordinates. To define all higher differentials uniformly and prove exactness, we rewrite the same free left R-modules in vertex coordinates. Think of $[g_1|\cdots|g_n]$ as n successive steps, with vertices

$$1, \quad g_1, \quad g_1g_2, \quad \ldots, \quad g_1\cdots g_n.$$

There are $n+1$ vertices. Multiplying the bar symbol by $a \in G$ translates every vertex on the left.

Let E_n be the free abelian group on ordered tuples $(x_0, \ldots, x_n) \in G^{n+1}$, with diagonal left G-action

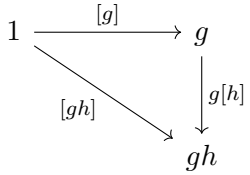
$$a(x_0, \ldots, x_n) = (ax_0, \ldots, ax_n).$$

This makes E_n a left R-module. The coordinate changes are

$$\begin{aligned}
a[g_1|\cdots|g_n] &\longleftrightarrow (a, ag_1, ag_1g_2, \ldots, ag_1\cdots g_n), \\
(x_0, \ldots, x_n) &\longleftrightarrow x_0[x_0^{-1}x_1|\cdots|x_{n-1}^{-1}x_n].
\end{aligned}$$

Here $a \in G$. The first vertex recovers $a = x_0$, and successive vertices recover $g_i = x_{i-1}^{-1}x_i$. Thus these are inverse bijections of the abelian-group bases. They commute with the left G-actions and extend additively to left R-linear isomorphisms $B_n^L \cong E_n$. In particular, the tuples starting at 1 form a free left R-basis of E_n. The entries g_i describe steps $x_i = x_{i-1}g_i$; the left R-action translates all vertices simultaneously.

For example, $[g|h]$ corresponds to $(1, g, gh)$, pictured as a formal triangle:



The edge labels are formal chains. The edge from g to gh is g times the edge from 1 to h, explaining its label $g[h]$. Reading the three edges with signs $+, -, +$ gives the relation $r(g, h)$ above.

The general differential: delete one vertex. The same rule applies in every degree: take the alternating sum of the tuples obtained by deleting one vertex. Define

$$\partial_n(x_0, \dots, x_n) = \sum_{i=0}^{n} (-1)^i (x_0, \dots, \widehat{x_i}, \dots, x_n), \qquad n \geq 1.$$

The hat means that the indicated entry is omitted. Deletion commutes with the left G-action, so these maps are left R-linear. In degree zero, the augmentation sends each (x_0) to 1.

Translate deletion of a vertex of $(1, g_1, g_1g_2, \dots, g_1 \cdots g_n)$ back into bar symbols:

- Deleting the first vertex leaves a tuple beginning at g_1. Factoring out its left translation gives $g_1[g_2|\cdots|g_n]$.
- Deleting the ith interior vertex joins two successive steps, replacing g_i, g_{i+1} by g_ig_{i+1}.
- Deleting the last vertex removes the last step g_n.

Thus the corresponding left R-linear differential is

$$\begin{aligned} d_n^L[g_1|\cdots|g_n] = {} & g_1[g_2|\cdots|g_n] \\ & + \sum_{i=1}^{n-1} (-1)^i [g_1|\cdots|g_ig_{i+1}|\cdots|g_n] \\ & + (-1)^n [g_1|\cdots|g_{n-1}]. \end{aligned}$$

For $n = 1, 2, 3$ this recovers the maps constructed above. We have now defined the augmented left bar sequence

$$\cdots \longrightarrow B_2^L \xrightarrow{d_2^L} B_1^L \xrightarrow{d_1^L} B_0^L \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0.$$

It remains to prove that the image of every differential is the whole kernel of the next one.

Proposition. The augmented left bar sequence is a free resolution of the trivial left R-module $\mathbb{Z}$.

Proof. Fix two vertex indices $i < j$. Deleting j and then i gives sign $(-1)^{i+j}$; deleting i and then the shifted index $j-1$ gives $(-1)^{i+j-1}$. The remaining tuples agree, so these terms cancel. Hence $\partial_{n-1}\partial_n = 0$ for $n \geq 2$. Also $\varepsilon\partial_1 = 0$. To prove exactness, define homomorphisms of abelian groups

$$\begin{aligned} & s_n : E_n \longrightarrow E_{n+1}, \quad s_n(x_0, \dots, x_n) = (1, x_0, \dots, x_n) \quad (n \geq 0), \\ & s_{-1} : \mathbb{Z} \longrightarrow E_0, \qquad\qquad s_{-1}(k) = k(1). \end{aligned}$$

Deleting the newly inserted first vertex gives the original tuple; all remaining terms cancel those in $s_{n-1}\partial_n$. Hence

$$\partial_{n+1}s_n + s_{n-1}\partial_n = \mathrm{id}_{E_n} \quad (n \geq 0), \qquad \varepsilon s_{-1} = \mathrm{id}_{\mathbb{Z}},$$

where $\partial_0 = \varepsilon$ in this augmented formula. In particular, a cycle z satisfies $z = \partial_{n+1}s_n(z)$, and the augmentation is surjective. The augmented sequence is therefore exact as a sequence of abelian

groups, hence also as a sequence of left R-modules. Each $E_n \cong B_n^L$ is free over R, as shown above. □

Remark. The maps s_n are $\mathbb{Z}$-linear, but generally not R-linear. For $a \in G$,

$$s_n(a(x_0, \dots, x_n)) = (1, ax_0, \dots, ax_n),$$
$$a\, s_n(x_0, \dots, x_n) = (a, ax_0, \dots, ax_n).$$

They prove exactness of the bar resolution, but do not give an R-linear contraction to which tensor or Hom over R could be applied. Thus this contraction does not force group homology or cohomology to vanish.

The right bar resolution. For homology with a coefficient left R-module, we need a right R-resolution of $\mathbb{Z}$. Use the same tuple groups E_n, now with the diagonal right action

$$(x_0, \dots, x_n)g = (x_0 g, \dots, x_n g).$$

Choose the representatives whose last vertex is 1:

$$[g_1|\cdots|g_n]_R \longleftrightarrow (g_1 \cdots g_n, g_2 \cdots g_n, \dots, g_n, 1).$$

Every tuple is uniquely a representative of this form multiplied on the right by x_n: here $g_i = x_{i-1}x_i^{-1}$. This gives the free right R-modules

$$B_n^R = \bigoplus_{(g_1,\dots,g_n)\in G^n} [g_1|\cdots|g_n]_R R.$$

The same alternating deletion boundary becomes

$$\begin{aligned} d_n^R[g_1|\cdots|g_n]_R = {} & [g_2|\cdots|g_n]_R \\ & + \sum_{i=1}^{n-1} (-1)^i [g_1|\cdots|g_i g_{i+1}|\cdots|g_n]_R \\ & + (-1)^n [g_1|\cdots|g_{n-1}]_R g_n. \end{aligned}$$

Here deleting the first vertex needs no coefficient, while deleting the last vertex leaves a tuple with last vertex g_n, which is moved to the coefficient on the right. In particular,

$$d_1^R[g]_R = [\,]_R - [\,]_R g.$$

The sign $1 - g$ comes from this choice of coordinates. With augmentation $\varepsilon([\,]_R r) = \varepsilon_R(r)$, this is a free right R-resolution of $\mathbb{Z}$.

Proof. Alternating deletion commutes with the diagonal right action, and the same abelian-group contraction s_n proves exactness. The unique representatives above give freeness. □

Normalization: an optional simplification. The preceding *unnormalized* bar resolutions already suffice for all our computations. Normalization is an optional way to remove redundancies caused by the identity element. For either the left or the right bar resolution,

$$d_1[1] = 0, \qquad d_2[1|1] = [1].$$

Thus [1] introduces a degree-one cycle that is already the boundary of [1|1]. Removing such symbols systematically in all degrees gives a smaller free resolution, as follows.

For either choice of side, let $D_n \subseteq B_n$ be the left or right R-submodule, respectively, generated by brackets $[g_1|\cdots|g_n]$ having some $g_i = 1$, and set $D_0 = 0$. Define

$$\overline{B}_n = B_n/D_n.$$

Equivalently, $\overline{B}_n^L$ is the free left R-module, and $\overline{B}_n^R$ the free right R-module, on brackets with $g_i \neq 1$ for every i. In degree zero the empty bracket remains. Use the same boundary formulas, interpreting any bracket containing 1 as zero. For example, $[g|g^{-1}]$ is allowed when $g \neq 1$, but the resulting term $[gg^{-1}] = [1]$ is zero.

Proposition. The augmented sequences $\overline{B}_\bullet^L \to \mathbb{Z}$ and $\overline{B}_\bullet^R \to \mathbb{Z}$ are free resolutions of the trivial left and right R-modules $\mathbb{Z}$, respectively.

Proof. In homogeneous coordinates, the discarded tuples are exactly those with two adjacent equal vertices. If $x_i = x_{i+1}$, the two boundary terms deleting x_i and x_{i+1} cancel. Every other boundary term still has adjacent equal vertices. Thus $D_\bullet$ is a subcomplex, and the boundary descends to the quotient. Inserting 1 at the beginning preserves any existing adjacent equality, so $s_n(D_n) \subseteq D_{n+1}$. The same augmented contraction therefore descends as maps of abelian groups and proves exactness of $\overline{B}_\bullet \to \mathbb{Z}$. Finally, D_n is spanned over R by a subset of the original basis, so the complementary brackets form a free basis of the quotient. □

For a finite group of order q, normalization reduces the number of free R-basis elements in degree $n \geq 1$ from q^n to $(q-1)^n$. For any group, it removes all brackets containing an identity entry from calculations. Both versions may be used, with the right resolution for homology and the left resolution for cohomology. Section 5.2 explains why the quotient maps preserve the resulting homology and cohomology: they are chain homotopy equivalences over R.

5.2 Explicit definitions of group homology and cohomology

Fix a left $R = \mathbb{Z}G$-module A. We first use the unnormalized free right resolution $B_\bullet^R \to \mathbb{Z}$ and free left resolution $B_\bullet^L \to \mathbb{Z}$. The balanced computation of Tor and the equivalent computation of Ext give

$$H_n(G;A) \cong H_n(B_\bullet^R \otimes_R A),$$
$$H^n(G;A) \cong H^n(\mathrm{Hom}_R(B_\bullet^L, A)).$$

In each case the augmentation term is omitted. We now write these complexes without referring to left or right R-linear maps on a resolution.

The bar chain complex. Put

$$C_n(G;A) := B_n^R \otimes_R A \cong \bigoplus_{G^n} A \qquad (n \geq 0).$$

An element is a finite sum of symbols $a[g_1|\cdots|g_n]$, where this notation means $[g_1|\cdots|g_n]_R \otimes a$ and is additive in $a \in A$. Since G^0 consists of one empty tuple, $C_0(G;A) = A$; we identify $a[]$

with a. For $n \geq 1$, the differential is

$$\begin{aligned}\partial_n(a[g_1|\cdots|g_n]) = a[g_2|\cdots|g_n] \\ + \sum_{i=1}^{n-1}(-1)^i a[g_1|\cdots|g_ig_{i+1}|\cdots|g_n] \\ + (-1)^n(g_na)[g_1|\cdots|g_{n-1}],\end{aligned}$$

and $\partial_0 = 0$. The last term follows from the tensor relation

$$([g_1|\cdots|g_{n-1}]_R g_n) \otimes a = [g_1|\cdots|g_{n-1}]_R \otimes (g_n a).$$

For example,

$$\begin{aligned}\partial_1(a[g]) &= a - ga, \\ \partial_2(a[g|h]) &= a[h] - a[gh] + (ha)[g].\end{aligned}$$

Thus

$$H_n(G;A) \cong \ker \partial_n / \operatorname{im} \partial_{n+1}.$$

In degree zero, $\operatorname{im}\partial_1$ is generated by the elements $a - ga$, so this gives $H_0(G;A) = A_G$, as required.

The bar cochain complex. An R-linear map $B_n^L \to A$ is determined by arbitrary values on its free basis. Consequently,

$$C^n(G;A) := \operatorname{Hom}_R(B_n^L, A) \cong \{f : G^n \to A\},$$

with pointwise addition. These are *all* functions, with no homomorphism condition or finite-support requirement. In degree zero we again have $C^0(G;A) = A$. The differential $\delta^n(f) = f \circ d_{n+1}^L$ is

$$\begin{aligned}(\delta^n f)(g_1,\ldots,g_{n+1}) = g_1 f(g_2,\ldots,g_{n+1}) \\ + \sum_{i=1}^{n}(-1)^i f(g_1,\ldots,g_ig_{i+1},\ldots,g_{n+1}) \\ + (-1)^{n+1} f(g_1,\ldots,g_n).\end{aligned}$$

Here $f(g_1,\ldots,g_n)$ denotes the value of the corresponding R-linear map on $[g_1|\cdots|g_n]_L$. The first term uses R-linearity: $f(g_1[\cdots]_L) = g_1 f([\cdots]_L)$. In low degrees,

$$\begin{aligned}(\delta^0 a)(g) &= ga - a, \\ (\delta^1 f)(g,h) &= gf(h) - f(gh) + f(g), \\ (\delta^2 f)(g,h,k) &= gf(h,k) - f(gh,k) + f(g,hk) - f(g,h).\end{aligned}$$

Hence

$$H^n(G;A) \cong \ker \delta^n / \operatorname{im} \delta^{n-1},$$

where $\delta^{-1} = 0$. In particular, $H^0(G;A) = \ker \delta^0 = A^G$.

The coefficient action appears at the *last* face in the chain formula and at the *first* face in the cochain formula. This follows from our right and left bar conventions, respectively; the formulas should not be interchanged for a general action.

For a 1-cochain $f : G \to A$, the cocycle condition is

$$f(gh) = f(g) + gf(h).$$

On the triangle above, this says that the value along $1 \to g \to gh$ agrees with the value along $1 \to gh$. Translating the second edge by g acts on its value by g as well. For trivial action, the condition reduces to $f(gh) = f(g) + f(h)$: precisely a group homomorphism from G to the additive group of A.

Homogeneous cochains. Using vertex coordinates, a degree-n cochain is an equivariant function $\Phi : G^{n+1} \to A$, meaning

$$\Phi(gx_0, \ldots, gx_n) = g\Phi(x_0, \ldots, x_n).$$

Its differential is simply alternating deletion:

$$(\delta^n \Phi)(x_0, \ldots, x_{n+1}) = \sum_{i=0}^{n+1} (-1)^i \Phi(x_0, \ldots, \widehat{x_i}, \ldots, x_{n+1}).$$

The correspondence with the functions $f : G^n \to A$ above is

$$\begin{aligned} \Phi(x_0, \ldots, x_n) &= x_0 f(x_0^{-1}x_1, \ldots, x_{n-1}^{-1}x_n), \\ f(g_1, \ldots, g_n) &= \Phi(1, g_1, g_1g_2, \ldots, g_1 \cdots g_n). \end{aligned}$$

These describe the same cochain complex: an R-linear map $E_n \to A$ is exactly an equivariant function on its tuple basis, extended additively. The coordinate isomorphisms $B_n^L \cong E_n$ give the two inverse formulas and identify the differentials. □

An R-linear coefficient map $u : A \to A'$ acts on chains by $a[g_1|\cdots|g_n] \mapsto u(a)[g_1|\cdots|g_n]$, and on cochains by $f \mapsto u \circ f$. These maps commute with the displayed differentials and induce the coefficient maps defined earlier.

Normalized chains and cochains. Using the normalized resolutions gives the smaller complexes

$$\begin{aligned} \overline{C}_n(G; A) &= \overline{B}_n^R \otimes_R A \cong \bigoplus_{(G\setminus\{1\})^n} A, \\ \overline{C}^n(G; A) &= \mathrm{Hom}_R(\overline{B}_n^L, A). \end{aligned}$$

The same boundary formula applies to normalized chains, with every symbol containing an entry 1 interpreted as zero. Normalized cochains are exactly the functions $f : G^n \to A$ that vanish whenever some argument is 1; their coboundary is the same formula as above. For example, a normalized 1-cochain satisfies $f(1) = 0$, and a normalized 2-cochain satisfies $f(1, g) = f(g, 1) = 0$. These conditions remove values on tuples containing the identity from the cochain calculation. For $n = 0$, both normalized terms are still A.

Proposition. The normalized complexes compute the same homology and cohomology:

$$\begin{aligned} H_n(G; A) &\cong H_n(\overline{C}_\bullet(G; A)), \\ H^n(G; A) &\cong H^n(\overline{C}^\bullet(G; A)). \end{aligned}$$

Proof. Both normalized bar complexes are free resolutions of the indicated trivial right and left R-modules, respectively. Thus the Tor and Ext computation theorems apply to them. More

specifically, the quotient maps $B_\bullet^L \to \overline{B}_\bullet^L$ and $B_\bullet^R \to \overline{B}_\bullet^R$ extend $\mathrm{id}_\mathbb{Z}$. By comparison and uniqueness of projective resolutions, these are chain homotopy equivalences over R. Tensoring the right-hand map with A, or applying $\mathrm{Hom}_R(-, A)$ to the left-hand map, therefore induces an isomorphism on homology or cohomology. In the Hom case, writing $q : B_\bullet^L \to \overline{B}_\bullet^L$ for the quotient, the map $\phi \mapsto \phi \circ q$ identifies the normalized cochains with the functions vanishing on degenerate tuples, as claimed. □

What the derived-functor description adds. We could take the displayed bar complexes and their homology or cohomology as the definitions of $H_n(G; A)$ and $H^n(G; A)$. No choice of resolution would then appear in the definition. The earlier theory explains why these constructions measure the failure of coinvariants to be exact and of invariants to be exact. It also proves that we may replace the bar resolution by any more convenient projective resolution, with the same results and the same maps on coefficients. Thus the bar formulas provide an explicit construction; the derived-functor description explains its relation to the module-theoretic operations from which we started.

5.3 A worked bar calculation

Return to $G = \langle t \rangle \cong \mathbb{Z}$ and $R = \mathbb{Z}[t, t^{-1}]$. Section 3.2 computed $H_n(G; \mathbb{Z})$ from a free resolution of the coefficient left R-module $\mathbb{Z}$, while Section 4.4 computed $H^n(G; \mathbb{Q})$ from an injective resolution of the coefficient left R-module $\mathbb{Q}$. Both coefficient actions were trivial. We now recover these same two calculations using the bar resolutions of the trivial right and left R-modules $\mathbb{Z}$, respectively. We choose the normalized versions to shorten the calculations; the unnormalized versions give the same results.

Normalized chains with integral coefficients. The normalized chain group in degree $n \geq 1$ is free abelian on

$$[t^{m_1} | \cdots | t^{m_n}], \qquad m_1, \ldots, m_n \in \mathbb{Z} \setminus \{0\}.$$

Thus there are infinitely many basis elements in every positive degree: normalization removes identity entries, but does not express every bracket as a multiple of $[t| \cdots |t]$. The empty bracket remains in degree zero, where $\overline{C}_0(G; \mathbb{Z}) = \mathbb{Z}$. Since the action is trivial, the first two boundaries are

$$\partial_1[t^m] = 0, \qquad \partial_2[t^p | t^q] = [t^q] - [t^{p+q}] + [t^p],$$

where $[t^0] = [1] = 0$ in degree one. For example,

$$\partial_2[t|t] = 2[t] - [t^2], \qquad \partial_2[t|t^{-1}] = [t^{-1}] + [t].$$

The relations $[t^2] = 2[t]$ and $[t^{-1}] = -[t]$ therefore hold in homology, although they are not equalities of degree-one chains.

Homology in degrees zero and one. The first boundary is zero, so $H_0(G; \mathbb{Z}) = \mathbb{Z}$. Every degree-one chain is a cycle, and the degree-two boundaries impose

$$[t^{p+q}] = [t^p] + [t^q] \qquad \text{in } H_1(G; \mathbb{Z}).$$

These relations also hold when $p = 0$ or $q = 0$, by $[t^0] = 0$. They imply $[t^m] = m[t]$ for every integer m. There is no further relation on $[t]$: the homomorphism

$$w : \overline{C}_1(G;\mathbb{Z}) \longrightarrow \mathbb{Z}, \qquad w\left(\sum_m a_m[t^m]\right) = \sum_m m a_m$$

vanishes on every boundary and sends $[t]$ to 1. Consequently $H_1(G;\mathbb{Z}) \cong \mathbb{Z}$, with generator represented by $[t]$. □

Cohomology in degrees zero and one, with rational coefficients. Now use the trivial left R-module $A = \mathbb{Q}$. A normalized 1-cochain is a function $f : G \to \mathbb{Q}$ satisfying $f(1) = 0$. The action is trivial, so $\delta^0 = 0$ and

$$(\delta^1 f)(t^p, t^q) = f(t^q) - f(t^{p+q}) + f(t^p).$$

Thus f is a cocycle exactly when $f(t^{p+q}) = f(t^p) + f(t^q)$, or equivalently

$$f(t^m) = mc \qquad (m \in \mathbb{Z})$$

for a single rational number $c = f(t)$. There are no nonzero 1-coboundaries because $\delta^0 = 0$. It follows that

$$H^0(G;\mathbb{Q}) = \mathbb{Q}, \qquad H^1(G;\mathbb{Q}) \xrightarrow{\cong} \mathbb{Q}, \quad [f] \longmapsto f(t),$$

with inverse $c \mapsto [t^m \mapsto mc]$. □

Higher degrees: comparison with short free resolutions. Both normalized bar complexes have nonzero terms in every degree. To show that no higher homology or cohomology survives, compare the bar resolutions with the following short resolutions of $\mathbb{Z}$:

$$0 \longrightarrow Re_1 \xrightarrow{d_1^L} Re_0 \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0, \quad d_1^L(e_1) = (t-1)e_0,$$
$$0 \longrightarrow e_1R \xrightarrow{d_1^R} e_0R \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0, \quad d_1^R(e_1) = e_0(1-t).$$

The first row consists of left R-modules and the second of right R-modules. Section 3.2 proved exactness of the first row. Since R is commutative, the same calculation applies on the right; changing $t-1$ to $1-t$ changes neither its kernel nor its image. Write $P_\bullet^L$ and $P_\bullet^R$ for the underlying complexes. They resolve the trivial left and right R-modules $\mathbb{Z}$, respectively; the coefficient left R-module for cohomology remains $\mathbb{Q}$.

Set

$$S_m = \begin{cases} 1 + t + \cdots + t^{m-1}, & m > 0, \\ 0, & m = 0, \\ -(t^m + t^{m+1} + \cdots + t^{-1}), & m < 0. \end{cases}$$

These Laurent polynomials satisfy

$$(t-1)S_m = t^m - 1, \qquad S_{p+q} = S_p + t^p S_q = S_q + S_p t^q, \qquad \varepsilon(S_m) = m.$$

For each side, define maps

$$i^L : P_\bullet^L \longrightarrow \overline{B}_\bullet^L, \qquad p^L : \overline{B}_\bullet^L \longrightarrow P_\bullet^L,$$
$$i^R : P_\bullet^R \longrightarrow \overline{B}_\bullet^R, \qquad p^R : \overline{B}_\bullet^R \longrightarrow P_\bullet^R.$$

On either side put $i_0(e_0) = [\,]$, $i_1(e_1) = [t]$, and $p_0([\,]) = e_0$, using the bar symbols of that side. In degree one set

$$p_1^L([t^m]_L) = S_m e_1, \qquad p_1^R([t^m]_R) = e_1 S_m.$$

Extend by left or right R-linearity as appropriate, and set all maps in degrees at least two to zero.

The identity $(t-1)S_m = t^m - 1$ verifies the degree-one equations on both sides. In degree two the equations are

$$p_1^L d_2^L [t^p|t^q]_L = (t^p S_q - S_{p+q} + S_p)e_1 = 0,$$
$$p_1^R d_2^R [t^p|t^q]_R = e_1(S_q - S_{p+q} + S_p t^q) = 0.$$

All higher equations are zero, and the equations for i^L, i^R follow from the displayed differentials of $[t]$. These are therefore chain maps extending $\mathrm{id}_{\mathbb{Z}}$. On each side $pi = \mathrm{id}_P$, since $S_1 = 1$. The comparison theorem gives $ip \simeq \mathrm{id}_{\overline{B}}$, with zero homotopy in degree -1, because both maps lift $\mathrm{id}_{\mathbb{Z}}$. Thus the short complexes and the corresponding bar complexes are chain homotopy equivalent over R. □

Apply tensor and Hom. For homology, the balanced computation theorem permits us to use the right resolution $P_\bullet^R \to \mathbb{Z}$. Tensoring i^R and p^R with the trivial left R-module $\mathbb{Z}$ gives a chain homotopy equivalence between $\overline{C}_\bullet(G;\mathbb{Z})$ and

$$0 \longrightarrow \underbrace{\mathbb{Z}}_{\text{degree 1}} \xrightarrow{0} \underbrace{\mathbb{Z}}_{\text{degree 0}} \longrightarrow 0.$$

In degree one the two maps are $1 \mapsto [t]$ and $[t^m] \mapsto m$, respectively, since $\varepsilon(S_m) = m$. Every bar cycle of degree at least two is therefore a boundary.

For cohomology, the equivalent computation theorem for Ext allows the left resolution $P_\bullet^L \to \mathbb{Z}$ to replace the injective resolution of $\mathbb{Q}$ used in Section 4.4. Explicitly, the maps

$$\mathrm{Hom}_R(\overline{B}_n^L, \mathbb{Q}) \longrightarrow \mathrm{Hom}_R(P_n^L, \mathbb{Q}), \quad \phi \longmapsto \phi \circ i_n^L,$$
$$\mathrm{Hom}_R(P_n^L, \mathbb{Q}) \longrightarrow \mathrm{Hom}_R(\overline{B}_n^L, \mathbb{Q}), \quad \psi \longmapsto \psi \circ p_n^L$$

give a cochain homotopy equivalence with

$$0 \longrightarrow \underbrace{\mathbb{Q}}_{\text{degree 0}} \xrightarrow{0} \underbrace{\mathbb{Q}}_{\text{degree 1}} \longrightarrow 0.$$

In degree one these maps send f to $f(t)$, and a rational number c to the function $t^m \mapsto mc$. Every bar cocycle of degree at least two is therefore a coboundary. Altogether,

$$H_n(G;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & n = 0, 1, \\ 0, & n \geq 2, \end{cases} \qquad H^n(G;\mathbb{Q}) \cong \begin{cases} \mathbb{Q}, & n = 0, 1, \\ 0, & n \geq 2. \end{cases}$$

These agree with the direct derived-functor calculations in Chapters 3 and 4. The bar resolution supplies explicit chains and cochains for every group; comparison with a shorter resolution explains the vanishing of the higher degrees here. □

A Existence of injective resolutions

We prove the existence statement used in Section 4.1. Throughout, R is a unital ring, not necessarily commutative, and all R-actions in this appendix are left actions.

A divisible target. An abelian group D is *divisible* if, for every $n \geq 1$ and $a \in D$, there exists $b \in D$ with $nb = a$.

Lemma. Every divisible abelian group is injective over $\mathbb{Z}$.

Proof. Let $B \subseteq C$ be abelian groups and $h : B \to D$ a homomorphism. Order its extensions $h_H : H \to D$, with $B \subseteq H \subseteq C$, by extension. A chain has an upper bound obtained by taking the union, so Zorn's lemma gives a maximal extension. If $H \neq C$, choose $x \in C \setminus H$ and write

$$\{k \in \mathbb{Z} : kx \in H\} = m\mathbb{Z}, \qquad m \geq 0.$$

If $m = 0$, choose $y = 0$; if $m > 0$, divisibility gives $y \in D$ with $my = h_H(mx)$. Define

$$h' : H + \mathbb{Z}x \longrightarrow D, \qquad h'(a + kx) = h_H(a) + ky.$$

To check well-definedness, suppose $a + kx = a' + k'x$. If $m = 0$, then $k = k'$ and $a = a'$. If $m > 0$, write $k - k' = \ell m$; then $a - a' = -\ell mx$, so

$$h_H(a) - h_H(a') + (k - k')y = -\ell h_H(mx) + \ell my = 0.$$

Thus h' is an additive extension to a strictly larger subgroup, contradicting maximality. Hence $H = C$, proving injectivity. □

In particular, $E = \mathbb{Q}/\mathbb{Z}$ is injective over $\mathbb{Z}$: it is divisible because, for any representative $q \in \mathbb{Q}$, the coset $q/n + \mathbb{Z}$ is an nth root of $q + \mathbb{Z}$. □

Embedding an abelian group. Every abelian group M embeds in an injective abelian group.

Proof. First, homomorphisms $M \to E$ separate points. For $0 \neq x \in M$ of finite order d, send x to $1/d + \mathbb{Z}$; if x has infinite order, send it to $1/2 + \mathbb{Z}$. Each prescription defines a homomorphism $\mathbb{Z}x \to E$ that is nonzero on x. Injectivity of E extends it to M. Set

$$S = \mathrm{Hom}_{\mathbb{Z}}(M, E), \qquad D = \prod_{\chi \in S} E, \qquad e : M \longrightarrow D, \quad e(x) = (\chi(x))_{\chi \in S}.$$

The product consists of all families, without a finite-support restriction. The map e is additive and injective by separation. To see that D is injective, let $B \subseteq C$ and $f : B \to D$ be additive. Extend each coordinate $f_\chi : B \to E$ to $\widetilde{f}_\chi : C \to E$ and assemble them into $c \mapsto (\widetilde{f}_\chi(c))_{\chi \in S}$. This is an additive extension of f to C. □

Proposition (Enough injectives). Every left R-module embeds into an injective left R-module.

Proof. Apply the preceding construction to the underlying abelian group of M, obtaining an additive injection $e : M \to D$, where D is injective over $\mathbb{Z}$. Define

$$J = \mathrm{Hom}_{\mathbb{Z}}(R, D), \qquad (r\phi)(s) = \phi(sr).$$

This is a left R-action: the distributive axioms follow from additivity, $1\phi = \phi$, and

$$(r_1(r_2\phi))(s) = \phi((sr_1)r_2) = ((r_1r_2)\phi)(s).$$

No R-action on D is required.

We first check that J is injective. Let $X \subseteq Y$ be left R-modules and $f : X \to J$ an R-linear map. The additive map $u : X \to D$, $u(x) = f(x)(1)$, extends to an additive map $v : Y \to D$. Put

$$\widetilde{f}(y)(s) = v(sy).$$

This is additive in both y and s, and

$$\widetilde{f}(ry)(s) = v(sry) = (r\widetilde{f}(y))(s).$$

Hence $\widetilde{f} : Y \to J$ is R-linear. For $x \in X$,

$$\widetilde{f}(x)(s) = u(sx) = f(sx)(1) = (sf(x))(1) = f(x)(s),$$

so it extends f.

Finally, define

$$\iota : M \longrightarrow J, \qquad \iota(x)(s) = e(sx).$$

The map ι is additive and R-linear, since $\iota(rx)(s) = e(srx) = (r\iota(x))(s)$. If $\iota(x) = 0$, evaluation at $s = 1$ gives $e(x) = 0$, hence $x = 0$. Thus ι is the required embedding into the injective left R-module J. □

Constructing the resolution. Choose an embedding $\iota : M \to I^0$ with I^0 injective, then embed the quotient $I^0/\iota(M)$ into an injective left R-module I^1. Define $d^0 : I^0 \to I^1$ as the quotient map followed by this embedding; its kernel is the image of M. Next embed $I^1/\operatorname{im} d^0$ into I^2 and define d^1 in the same way. Continuing gives

$$0 \longrightarrow M \longrightarrow I^0 \xrightarrow{d^0} I^1 \xrightarrow{d^1} I^2 \longrightarrow \cdots,$$

with $\ker d^{n+1} = \operatorname{im} d^n$ at every step. This is an injective resolution of M. □